

Early-life stress induces persistent astrocyte dysfunction associated with fear generalisation

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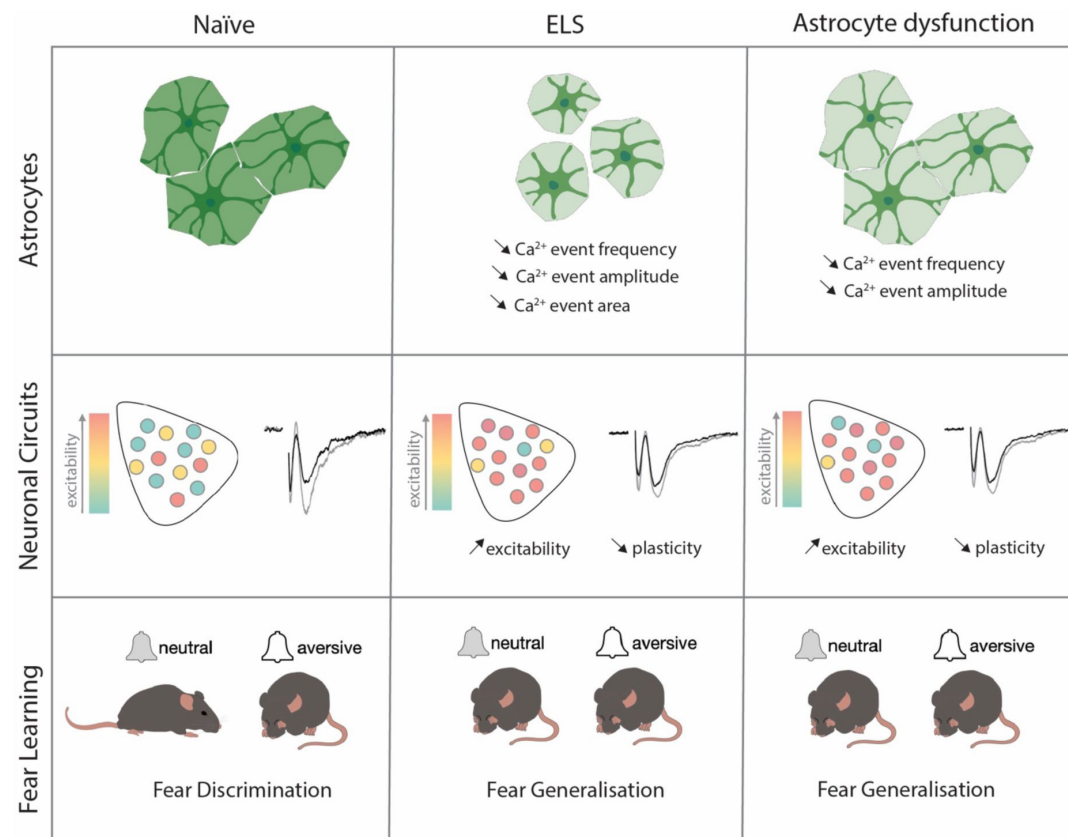
This **important** paper explores the impact of early life stress (ELS) on adult brain and behavior. The significance of the **convincing** findings are that they implicate regulation of non-neuronal cells in the development of brain and behavioral dysfunction associated with ELS. With an elegant combination of behavioral models, morphological and functional assessments using immunostaining, electrophysiology, and viral-mediated loss-of-function approaches, the authors report that astrocyte dysfunction plays a role in ELS responses. The work is of interest to a broad behavioral and cellular neuroscience audience.

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Abstract

Early-life stress can have lifelong consequences, enhancing stress susceptibility and resulting in behavioural and cognitive deficits. While the effects of early-life stress on neuronal function have been well-described, we still know very little about the contribution of non-neuronal brain cells. Investigating the complex interactions between distinct brain cell types is critical to fully understand how cellular changes manifest as behavioural deficits following early-life stress. Here, using male and female mice we report that early-life stress induces anxiety-like behaviour and fear generalisation in an amygdala-dependent learning and memory task. These behavioural changes were associated with impaired synaptic plasticity, increased neural excitability, and astrocyte hypofunction. Genetic perturbation of amygdala astrocyte function by either reducing astrocyte calcium activity or reducing astrocyte network function was sufficient to replicate cellular, synaptic, and fear memory generalisation associated with early-life stress. Our data reveal a role of astrocytes in tuning emotionally salient memory and provide mechanistic links between early-life stress, astrocyte hypofunction, and behavioural deficits.

Summary Figure



Introduction

Both human and rodent studies have provided compelling evidence for the long-lasting effects of adverse early-life experiences, highlighting enhanced susceptibility to subsequent stressors later in life (1–5). Early-life stress (ELS) has widespread effects across the brain and can influence many neural circuits including those involved in threat detection, emotion, cognitive processing, and reward seeking behaviours (6–9). Research investigating the neurobiology of stress has predominantly focused upon the effects on neurons and resulting behaviour (10–19). We have gleaned tremendous insight into disorders of stress from these studies, yet the links between stress, brain circuits and behaviour remain tenuous. Within this framework, the contribution of distinct brain cells, including astrocytes, remains largely untested. Glial cells, which comprise approximately half the cell population in the brain have been shown to directly regulate synaptic transmission and plasticity (20–29), thereby influencing many discrete behaviours (30–40). Despite research suggesting important roles for astrocytes in regulating affective states (31,33), fear (33), reward and addiction behaviours (36,41,42), whether astrocyte dysfunction prompts stress-induced behavioural impairments remains poorly understood.

First, we characterised whether ELS influenced stress hormone levels across our experimental paradigm from pre-stress until young adulthood. We report that that ELS induces a transient increase in blood glucocorticoids that peaks during late adolescence/young adulthood. Turning to behaviour, this increase in glucocorticoid levels was associated with anxiety-like behaviour in the elevated plus maze.

Next, we set out to determine whether ELS influenced cognitive function using an amygdala-dependent threat discrimination assay. This task was specifically chosen to test the effects of ELS on amygdala function as this brain region is involved with threat detection and associative learning that is acutely sensitive to stress (17,18,43–45). We employed an auditory discriminative fear conditioning paradigm that requires synaptic plasticity changes in the lateral amygdala (46) and found that ELS impairs discriminative fear memory resulting in fear generalisation. These hormonal and behavioural changes were associated with impaired synaptic plasticity and enhanced excitability of cells in the lateral amygdala, evidenced by increased c-Fos labelling.

Finally, we investigated the impact of ELS on astrocytes in the lateral amygdala revealing changes in specific proteins associated with morphological reorganisation and astrocyte network function. ELS resulted in decreased frequency and amplitude in astrocyte calcium activity. To directly implicate astrocytes with behavioural dysfunction we used two distinct viral approaches to mimic stress-induced astrocyte dysfunction specifically in the lateral amygdala. Impairment of astrocyte network function, or decreasing calcium activity, recapitulated ELS-induced fear generalisation, synaptic, and excitability phenotypes supporting the hypothesis that astrocytes play an influential role in the effects of stress on neural circuits and behaviour. Together, these findings identify astrocytes as central elements regulating amygdala-dependent affective memory and highlight astrocytes as putative targets for the treatment of stress disorders.

Results

Early-life stress increases blood corticosterone and induces anxiety-like behaviours

We employed an ELS paradigm which combines limited bedding and nesting materials as well as maternal separation from postnatal days 10 to 17 (Figure 1A) and has been shown to induce life-long stress susceptibility (6). We first set out to determine the impact of this ELS paradigm on stress hormone levels across our experimental window. Taking trunk blood from mice across several litters at different time points in the ELS paradigm: pre-stress (P10), end of ELS paradigm (P17), at the start (P45) and end of our experimental window (P70), we report that this ELS paradigm induces a latent increase in blood corticosterone with a significant increase occurring only in late adolescence P45, long after the termination of the stress paradigm (Figure 1B).

Next, we investigated whether this ELS paradigm results in long-term behavioural change, beginning with anxiety-like behaviours. We found that while there were no changes in behaviour in the open-field task (Figure 1D, E), we did see an increase in locomotor activity, i.e. distance travelled, following ELS (Figure 1F). Using the elevated plus maze we report increased anxiety-like behaviour with an increase in time spent in the closed arms and a reduction in time spent in the open arms of the maze (Figure 1H, I). This was also accompanied by an increase in distance travelled within the maze (Figure 1J), similar to our findings in the open field. Increased locomotion in response to stress is consistent with previous reports (15,47).

Amygdala-dependent memory, plasticity, and excitability are disrupted by early-life stress

Considering the strong association between stress disorders, anxiety, and amygdala dysfunction, we next set out to determine whether amygdala-dependent behaviour was impacted by ELS. We specifically targeted lateral amygdala-dependent learning and memory using an auditory discriminative fear conditioning paradigm. In brief, during this behavioural task mice were first exposed to a neutral auditory cue (conditioned stimulus -; CS-) followed by a distinct auditory cue (CS+) paired with footshock (unconditioned stimulus; US). Twenty-four hours later mice were

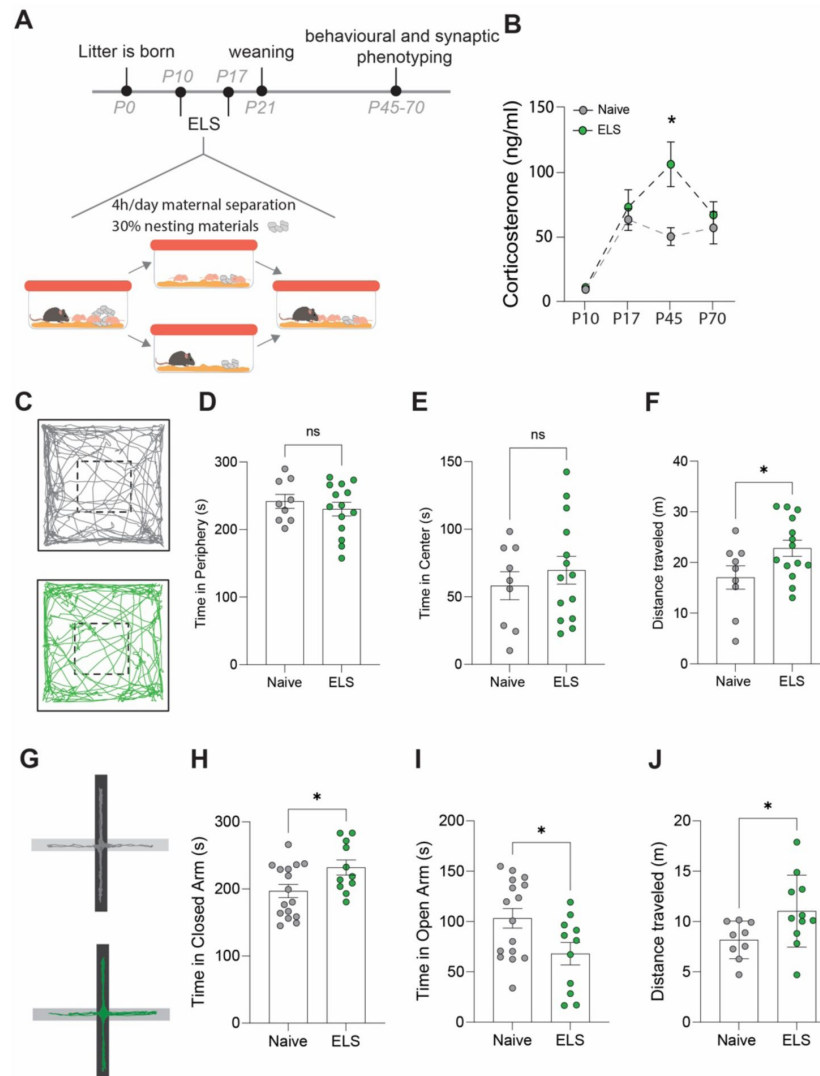


Figure 1.

Effects of ELS on anxiety-like behaviour.

(A) Timeline of ELS and behavioural assays. **(B)** Serum corticosterone levels are increased during adolescence, following ELS. $P=0.01$, Naive $N=9$, ELS $N=11$ mice. **(C)** Representative movement tracks of naive (grey) and ELS (green) mice in open field task. **(D)** No significant difference in time spent in the periphery of open field maze following ELS. $P=0.4591$, Naive $N=9$, ELS $N=14$ mice. **(E)** No significant difference in time spent in the center of open field maze following ELS. $P=0.4591$, Naive $N=9$, ELS $N=14$ mice. **(F)** Total distance travelled during open field test was increased in ELS mice. $P=0.04$, Naive $N=9$, ELS $N=14$ mice. **(G)** Representative locomotor tracks in elevated plus maze in naive and ELS mice, black segment represents closed arms with grey arms representing open arms. **(H)** ELS mice had increased time spent in closed arms $P=0.0276$, Naive $N=18$, ELS $N=11$ mice. **(I)** ELS mice had decreased time spent in open arms $P=0.0276$, Naive $N=18$, ELS $N=11$ mice. **(J)** Total distance travelled during elevated plus maze test was increased by ELS. $P=0.02$, Naive $N=9$, ELS $N=11$ mice. See Table S1 for detailed statistical summaries.

placed in a novel environment to remove contextual triggers and exposed to auditory cues from previous day without footshock (**Figure 2B** [↗](#)). Memory precision was measured using percent time freezing during presentation of neutral (CS-) and conditioned (CS+) auditory cues. We report that mice accurately learn this task and can readily distinguish between neutral and conditioned cues (**Table S1** [↗](#)).

Using this paradigm, we observed no effect of stress on freezing responses to the CS-presentations during conditioning (**Figure 2C, grey area** [↗](#)), ruling out ELS-induced hypersensitivity or hypervigilance to novel auditory cues. We also found that fear acquisition, evidenced by freezing to the CS+/US pairing, was unaffected by ELS (**Figure 2C** [↗](#)). In contrast, memory recall was affected with ELS mice exhibiting enhanced fear responses to neutral cues (CS-; **Figure 2D, grey area** [↗](#)), with no difference in freezing to conditioned cues (CS+; **Figure 2D** [↗](#)) compared to naïve mice. Additional analysis comparing average freezing responses to all tone presentations of CS- and CS+ supported these observations with no differences between naïve and ELS during conditioning (**Figure 2 – figure supplement 2A** [↗](#)); while during memory recall ELS mice showed enhanced freezing to CS-presentations compared to naïve counterparts with no difference in freezing to CS+ (**Figure 2 – figure supplement 2B** [↗](#)).

ELS-induced enhancement of fear responses to neutral cues is indicative of fear generalisation, a common stress-related behavioural phenotype conserved between rodents and humans ([48](#) [↗](#), [49](#) [↗](#)). This behavioural observation is more clearly depicted using a discrimination index (DI) that takes into account freezing responses to neutral and conditioned cues. Using this measure, $DI > 0$ reflects accurate discrimination, $DI = 0$ no discrimination and $DI < 0$ reflects a discrimination error, *i.e.* higher freezing to the neutral than conditioned cue. Using this index, we confirm a decrease in fear discrimination in ELS mice compared to naïve mice (**Figure 2E** [↗](#)). Together these data reveal long-term behavioral changes with ELS enhancing affective responses resulting in fear generalisation in adult mice.

Investigation of potential sex effects revealed no differences between naïve male and female mice in this paradigm with both sexes showing equal performance in learning (**Figure 2 – figure supplement 1A** [↗](#)), memory (**Figure 2 – figure supplement 1B** [↗](#)), and auditory discrimination (**Figure 2 – figure supplement 1C** [↗](#)). In ELS mice, we report a sex-dependent effect on fear acquisition with female mice exhibiting increased freezing responses to CS+/US pairing compared to males (**Figure 2 – figure supplement 1D** [↗](#)). Despite this, both sexes show equal performance during memory recall and auditory discrimination (**Figure 2 – figure supplement 1E, F** [↗](#)).

Auditory fear conditioning paradigms have been correlated with the potentiation of post-synaptic responses to cortical auditory afferents, in the lateral amygdala ([46](#) [↗](#)). As such, we set out to determine whether ELS impacts synaptic plasticity in cortico-amygdala circuits using acute brain slice electrophysiology and recorded field excitatory postsynaptic potentials (fEPSPs) by stimulating cortical inputs from the external capsule (**Figure 2F** [↗](#)). We report that ELS impaired long-term potentiation (LTP) of synaptic transmission in this cortico-amygdala circuit (**Figures 2G, H** [↗](#) and **Figure 2 – figure supplement 2C** [↗](#)). This synaptic observation is consistent with findings using distinct stress paradigms ([20](#) [↗](#)), and highlights the important influence of stress on synaptic function across various brain regions.

Fear learning and memory in the lateral amygdala has also been correlated with the allocation of a small number of neurons to a fear engram ([50](#) [↗](#)). These engrams are thought to be a possible neural substrate for memory and their formation depends on neural excitability levels at the moment of acquisition ([51](#) [↗](#), [52](#) [↗](#)). As both glucocorticoid injections or acute stress have been shown to increase engram size and enhance fear generalisation ([53](#) [↗](#), [54](#) [↗](#)), we set out to determine whether ELS results in memory changes due to alterations in putative engram recruitment. We collected brain tissue 90 minutes after conditioning (**Figure 2I** [↗](#)) and carried out immunostaining for the immediate early gene c-Fos, a commonly used activity marker, in the

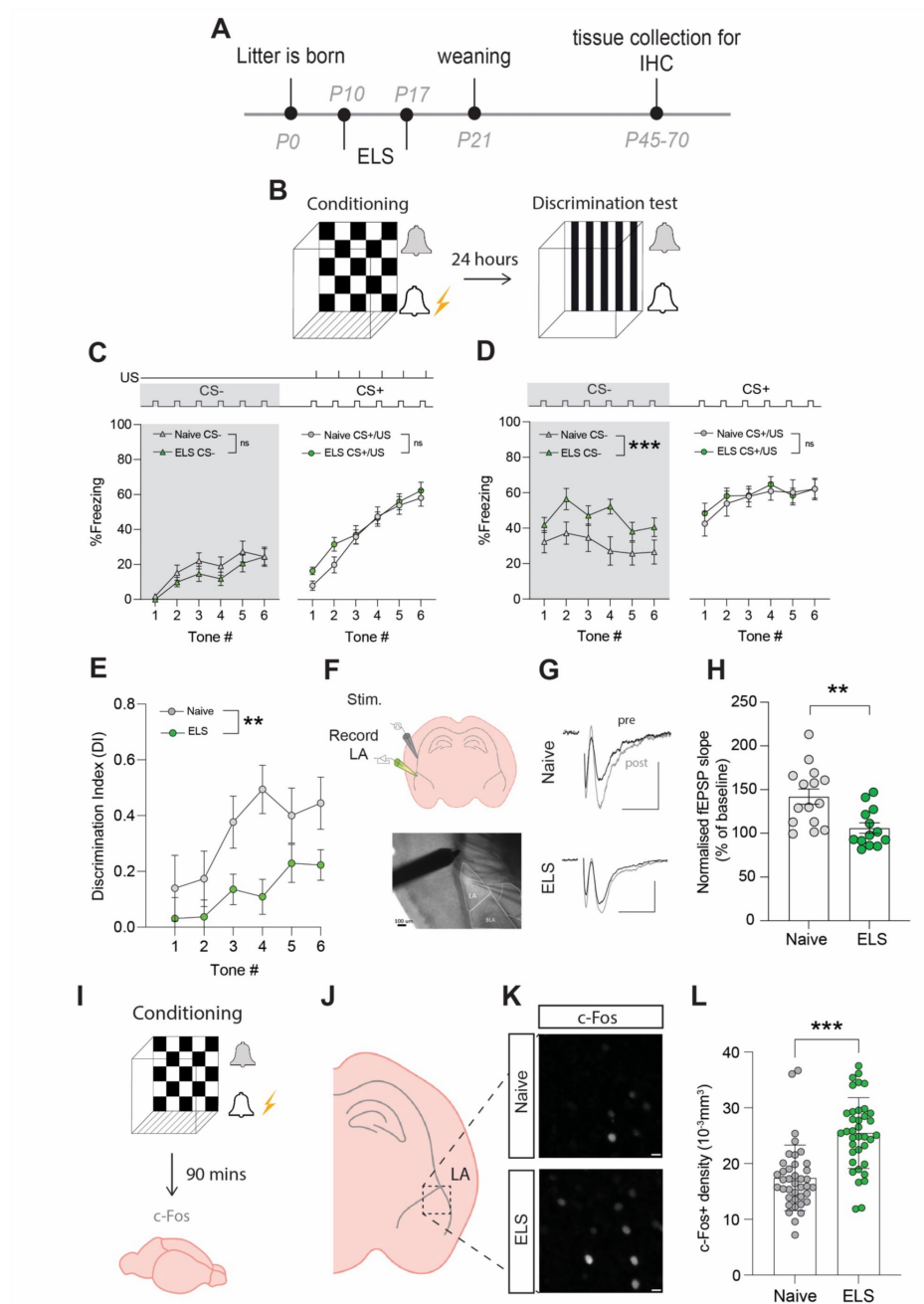


Figure 2.

ELS alters fear memory, synaptic plasticity, and neural excitability.

(A) Experimental timeline. **(B)** Timeline of auditory discriminative fear conditioning paradigm. **(C)** ELS did not affect % Freezing to neutral and aversive auditory cues during learning. Naïve N=20, ELS N=17. **(D)** % time freezing during presentation of neutral auditory cue (CS-; $p=0.0002$) and aversive cue (CS+; $p=0.99$). Naïve N=20, ELS N=17. **(E)** Discrimination Index was impaired by ELS. $P=0.004$. Naïve N=20, ELS N=17. **(F)** Schematic representation of electrode placement in slice electrophysiology experiment (top) with DIC image below. **(G)** Representative EPSP traces from naïve (top) and ELS (bottom) brain slices during baseline (black trace) and after LTP stimulation (grey trace). Scale bars = 0.5mV and 10ms. **(H)** LTP was impaired by ELS. $P=0.002$. Naïve N=15, ELS N=13. **(I)** Diagram of procedure for c-Fos staining experiments. **(J)** Brain slices containing lateral amygdala (LA) were selected for c-Fos staining. **(L)** Region of interest extracted from full amygdala image. Scale bars = 10 μ m. **(K)** c-Fos positive cell density was increased by ELS. $P<0.0001$. Naïve n=39 slices N=10 mice, ELS n=37 slices N=10 mice. See Table S1 for detailed statistical summaries.

lateral amygdala (**Figures 2J, K, L** [↗](#) and **Figure 2 – figure supplement 2D, E** [↗](#)). Fear acquisition in ELS mice resulted in a significant increase in c-Fos density, suggestive of increased excitability levels. These results are consistent with reports of c-Fos density inversely correlating with memory accuracy ([52](#) [↗](#)). Further investigation of potential sex differences revealed that while there were no differences in c-Fos densities between naïve female and male animals (**Figure 2 – figure supplement 2F** [↗](#)), we observed a significantly higher c-Fos density in ELS male mice compared to females following fear acquisition (**Figure 2 – figure supplement 2G** [↗](#)).

Recently, astrocytes have been reported to be recruited to fear engrams during a hippocampus-dependent learning and memory task ([55](#) [↗](#)). As such, we quantified the percentage of astrocytes expressing the activity marker c-Fos following fear conditioning in mice and report similar values to those found in the hippocampus (**Figure 2 – figure supplement 3A, B** [↗](#)) suggestive of modest recruitment of astrocytes in amygdala-dependent memory engrams.

Early-life stress induces persistent changes in astrocytes

To determine the influence of ELS on astrocytes in the lateral amygdala we carried out immunostaining on brain slices taken from a separate cohort of naïve and ELS mice. We first investigated whether increased blood corticosterone affects glucocorticoid receptor (GR) expression and localisation in astrocytes. Using immunostaining, we confirmed that this hormone receptor is abundantly expressed in lateral amygdala astrocytes (**Figure 3B** [↗](#)). We further investigated the subcellular localisation of GR within astrocytes. When inactivated GRs are restricted to the cytosol, upon binding to agonists such as corticosterone however, GRs can translocate to the nucleus to influence gene expression ([56](#) [↗](#)). Calculating a ratio of cytosolic vs nuclear GR in astrocytes as a proxy for activity, we found an increase in the nuclear fraction of GR in astrocytes following ELS (**Figure 3C** [↗](#)), suggestive of increased activation of glucocorticoid signaling in lateral amygdala astrocytes following ELS.

To assess putative changes in astrocyte structure we investigated expression of the intermediate filament protein glial fibrillary acidic protein (GFAP; **Figure 3D, E** [↗](#)) and found a significant decrease in normalised GFAP fluorescence intensity in the lateral amygdala following ELS (**Figure 3F** [↗](#)). While functional consequences of GFAP changes are difficult to define, we interpret these modifications in GFAP expression to be indicative of an astrocytic response to stress. We next targeted astrocyte-specific proteins with known roles in influencing synaptic function including Cx43, a gap junction protein that composes astrocyte networks ([57](#) [↗](#)) and the glutamate transporter GLT-1, responsible for the efficient clearance of glutamate from the synaptic cleft ([41](#); **Figure 3D, E** [↗](#)).

We observed a decrease in fluorescence intensity for the gap junction protein Cx43 (**Figure 3G** [↗](#)) while expression of the glutamate transporter GLT-1 remained unaffected by ELS (**Figure 3H** [↗](#)). We did not observe any significant sex differences in the expression of GFAP or Cx43 in control or ELS conditions, with similar expression levels in naïve and the same effect of stress on these proteins in both male and female mice (**Figure 3 – figure supplement 1A, B** [↗](#)). These data suggest that astrocytes mount a specific response to ELS which could result in discrete changes in astrocyte network function such as the uncoupling of the astrocytic Cx43 gap junction networks in the lateral amygdala.

Early-life stress impairs astrocyte calcium activity

While the ELS-induced changes in astrocyte protein expression are indicative of functional modifications, this cannot be assumed and needs to be tested. We chose to assess potential stress-induced changes in astrocyte calcium activity employing the membrane-tethered calcium sensor lck-GCaMP6f. Lateral amygdala astrocytes were specifically targeted using local injection of viral constructs with GCaMP6f expression driven by the GfaABC1D promoter. Two-photon laser scanning microscopy was used in acute brain slices a minimum of two weeks following virus

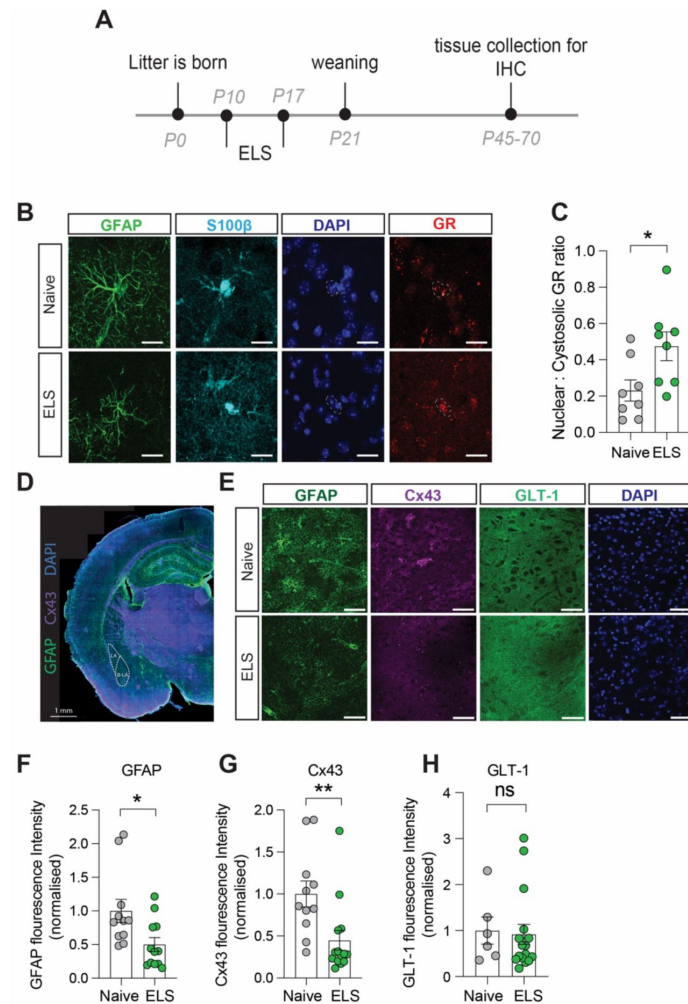


Figure 3.

Long-term astrocyte dysfunction after early-life stress.

(A) Experimental timeline. **(B)** Representative immunostaining of GFAP, S100 β , GR, DAPI, and GR/DAPI merge in Naïve and ELS conditions. Scale bars = 20 μ m **(C)** Nuclear/cystolic GR ratio in astrocytes was increased after ELS. $P=0.026$. Naïve $N=8$, ELS $N=8$. **(D)** Representative slide scan image of immunostaining for astrocyte proteins GFAP and Cx43. **(E)** Representative immunostaining of GFAP, Cx43, GLT-1, and DAPI in naïve and ELS conditions. Scale bars = 50 μ m **(F)** GFAP staining was reduced in ELS mice. $P=0.02$, Naïve $N=11$, ELS $N=12$. **(G)** Cx43 staining was decreased in ELS mice. $P=0.008$. Naïve $N=11$, ELS=14. **(H)** GLT-1 expression was unchanged by ELS. $P=0.8$. Naïve $N=6$, ELS $N=16$. See Table S1 for detailed statistical summaries.

injection (**Figure 4A**) allowing for quantification of dynamic elevations in intracellular calcium in astrocytic processes in naïve (**Figure 4B, C**) and ELS mice (**Figure 4D, E**). Data was analysed using the AQUA2 pipeline allowing for accurate and unbiased analysis of spatial and temporal dynamics of astrocyte calcium activity (59).

We report that ELS results in astrocyte calcium hypofunction, evidenced by decreased calcium event frequency (**Figure 4F, G**), amplitude (**Figure 4H, I**) and size of individual events (**Figure 4J, K**). Additionally, we observed a modest increase in rise time of calcium transients (**Figure 4L, M**) with no significant change in event duration (**Figure 4 N, O**) or decay time (**Figure 4P, Q**). Together, these results are suggestive of astrocyte hypofunction following ELS.

Unexpectedly, our experiments revealed sexual dimorphism in astrocytic calcium dynamics in the lateral amygdala between naïve female and male mice. Specifically, we found that naïve female mice exhibited larger event amplitudes (**Figure 4 – figure supplement 1A, B**), event area (**Figure 4 – figure supplement 1D, E**) and rise time (**Figure 4 – figure supplement 1G, H**) compared to naïve male mice. Interestingly, all sex differences were suppressed by stress with ELS-induced decrease in astrocyte activity normalising all parameters between sexes (**Figure 4 – figure supplement 1A, C, D, E, G, I**). No sex differences were observed for event frequency (**Figure 4 – figure supplement 1J-L**), duration (**Figure 4 – figure supplement 1M-O**) or decay time (**Figure 4 – figure supplement 1P-R**) in any conditions.

Astrocyte dysfunction alone phenocopies the effects of ELS on fear generalisation, synaptic plasticity, and excitability

To implicate astrocytes in the cellular and behavioural effects of ELS we carried out a series of genetic loss-of-function experiments, specifically targeting astrocytes in the lateral amygdala using viral approaches (**Figure 5A, B, C** and **Figure 5 – figure supplement 1A**). Based on the putative astrocyte network dysfunction we identified in fixed tissue (**Figure 3G**), we targeted astrocyte network function by overexpressing a dominant negative Cx43 (dnCx43). We have previously shown this manipulation to completely occlude functional coupling between neighbouring astrocytes by introducing a point mutation that blocks the pore of gap junction channels (dnCx43) (20). Second, to mimic the impact of ELS on astrocyte calcium activity (**Figure 4**), we used a calcium extruder pump (hPMCA2/b, a.k.a. CalEx) (60,61) that we validated to show that expression of this pump in lateral amygdala astrocytes (**Figure 5 – figure supplement 2A-E**) effectively reduces frequency (**Figure 5 – figure supplement 2F**) and amplitude (**Figure 5 – figure supplement 2G**) of astrocyte calcium events without affecting event size, duration, rise time, or decay (**Figure 5 – figure supplement 2H-K**). This manipulation closely mimics the effects of ELS on astrocyte calcium activity, allowing us to parse the precise contribution of astrocyte activity in behaviour.

Initially, we screened for changes in anxiety-like behaviours using the elevated plus maze and report no impact of lateral amygdala-specific astrocyte manipulation on the time spent in the open versus closed arms or in total distance traveled (**Figure 5 – figure supplement 3A-C**) nor in time spent in the periphery versus the middle of the open field maze or in total distance traveled (**Figure 5 – figure supplement 3D-F**). Next, using auditory discriminative fear conditioning paradigm, we found that learning was unaffected by astrocyte dysfunction (**Figure 5D**, **Figure 5 – figure supplement 4A**, and **Table S1**). We also report that similar to ELS, lateral amygdala astrocytic dysfunction does not lead to hypervigilance to neutral auditory cues (**Figure 5D**, grey area). During memory recall, however, we found that both astrocyte hypofunction manipulations enhanced fear responses to neutral cues, without affecting freezing to conditioned cues (**Figure 5E** and **Figure 5 – figure supplement 4B**). This behavioural phenotype induced by astrocyte dysfunction alone, with either dnCx43 or CalEx, mimicked the effects of ELS on auditory fear discrimination, i.e. enhanced fear generalisation. Accordingly, quantification of discrimination index revealed an impairment in discriminative fear memory in both dnCx43 and CalEx

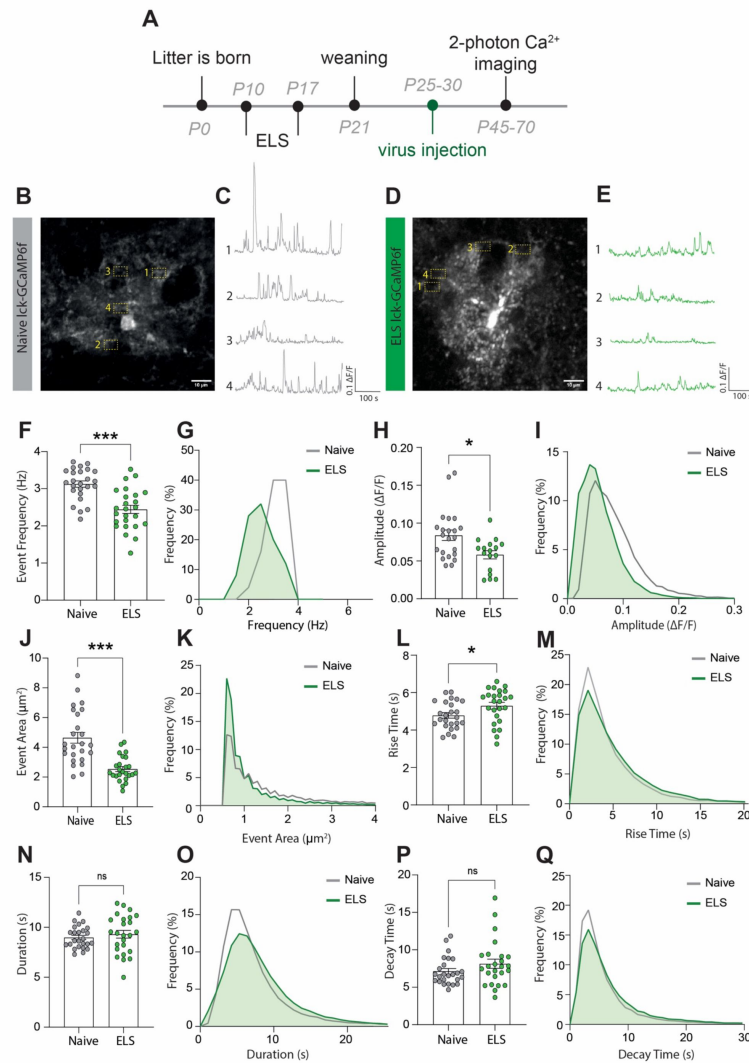


Figure 4.

ELS induces a hypofunction in lateral amygdala astrocyte calcium signaling.

(A) Experimental timeline including age of viral injection and experimental window for two-photon calcium imaging. **(B)** Representative image of a lck-GCaMP6f-expressing astrocyte in the lateral amygdala with 4 regions of interest from a naive animal. Scale bar = 10 μm . **(C)** Representative calcium-signaling traces from 4 regions of interest shown in **B**. **(D)** Representative image of a lck-GCaMP6f-expressing astrocyte in the lateral amygdala with 4 regions of interest from an ELS animal. Scale bar = 10 μm . **(E)** Representative calcium-signaling traces from 4 regions of interest shown in **D**. **(F)** Frequency of calcium events was decreased lateral amygdala astrocytes after ELS, $P < 0.0001$, Naive $n = 25$ astrocytes from $N = 8$ animals, ELS $n = 25$ astrocytes from $N = 7$ animals. **(G)** Frequency distribution of calcium event frequency from naive (grey line) and ELS (green line) animals. **(H)** Amplitude of calcium events was decreased after ELS, $P = 0.0165$, Naive $n = 22$ astrocytes from $N = 8$ animals, ELS $n = 17$ astrocytes from $N = 7$ animals. **(I)** Frequency distribution of calcium event amplitude from naive (grey line) and ELS (green line) animals. **(J)** Area of calcium events was decreased after ELS, $P < 0.0001$, Naive $n = 25$ astrocytes from $N = 8$ animals, ELS $n = 25$ astrocytes from $N = 7$ animals. **(K)** Frequency distribution of event area from naive (grey line) and ELS (green line) animals. **(L)** Rise time of calcium events was increased following ELS, $P = 0.0311$, Naive $n = 25$ astrocytes from $N = 8$ animals, ELS $n = 25$ astrocytes from $N = 7$ animals. **(M)** Frequency distribution of calcium event rise from naive (grey line) and ELS (green line) animals. **(N)** Calcium event duration was unchanged by ELS, $P = 0.4740$, Naive $n = 25$ astrocytes from $N = 8$ animals, ELS $n = 25$ astrocytes from $N = 7$ animals. **(O)** Frequency distribution of calcium event duration from naive (grey line) and ELS (green line) animals. **(P)** Decay time of calcium events was unaltered by ELS, $P = 0.2164$, Naive $n = 25$ astrocytes from $N = 8$ animals, ELS $n = 25$ astrocytes from $N = 7$ animals. **(Q)** Frequency distribution of calcium event decay time duration from naive (grey line) and ELS (green line) animals.

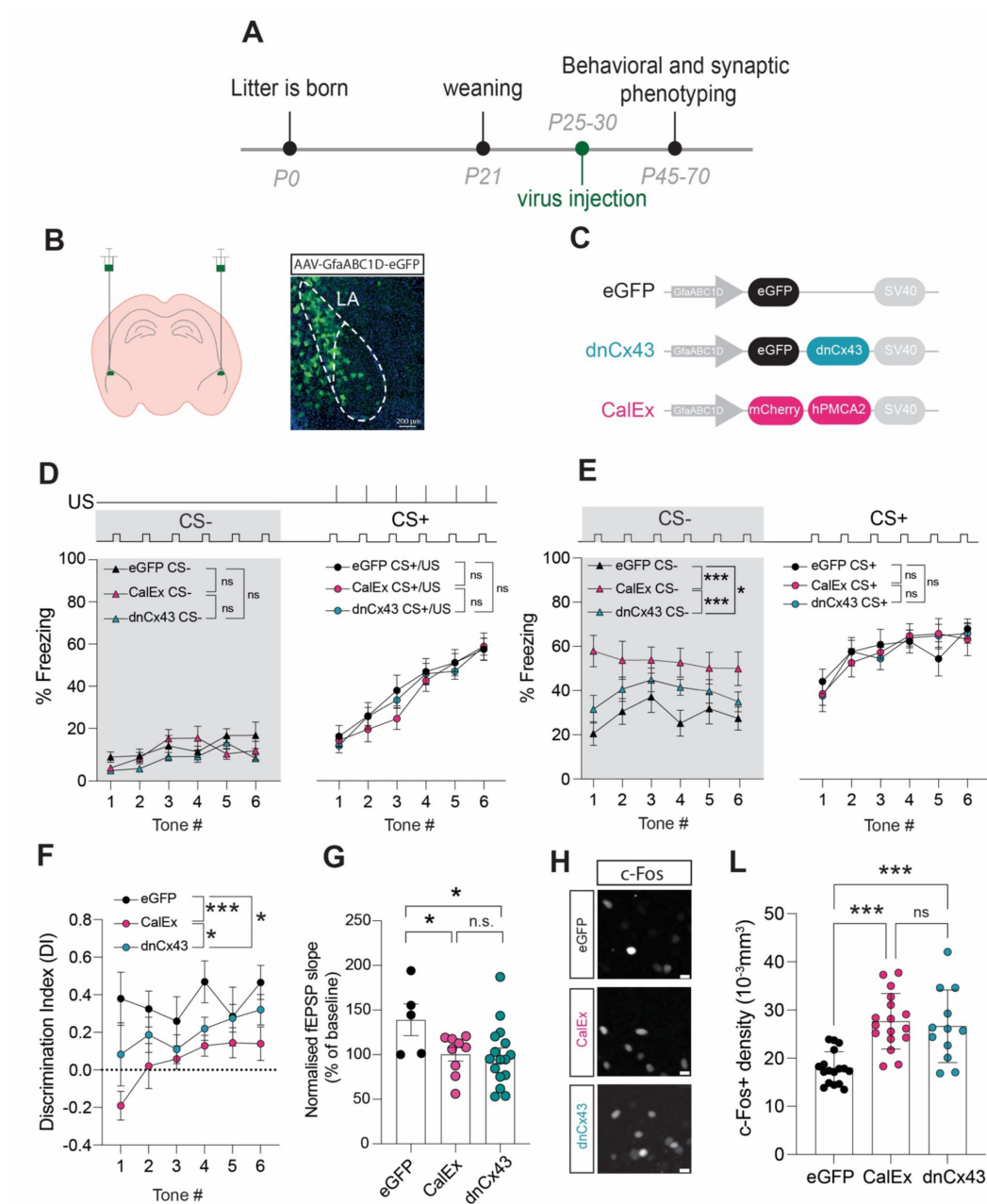


Figure 5.

Astrocyte dysfunction mimics the effects of ELS on behaviour, synapses, and neural excitability.

(A) Representative image showing localisation of viral vector in lateral amygdala. **(B)** Viral constructs used to manipulate astrocyte function. **(C)** Astrocyte dysfunction did not affect % Freezing to neutral and aversive auditory cues during learning. eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(D)** % time freezing during presentation of neutral auditory cue (CS-) was increased with astrocyte dysfunction (eGFP vs CalEx: $P<0.0001$; eGFP vs dnCx43: $P=0.029$; CalEx vs dnCx43: $P=0.0005$) with no impact on % Freezing to aversive cue (CS+). eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(E)** Discrimination Indexes were impaired by astrocyte dysfunction. eGFP vs CalEx: $P<0.0001$; eGFP vs dnCx43: $P=0.013$; CalEx vs dnCx43: $P=0.03$. eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(F)** Astrocyte dysfunction occluded the induction of LTP. eGFP vs CalEx: $P=0.04$; eGFP vs dnCx43: $P=0.02$; CalEx vs dnCx43: $P=0.88$. eGFP N=5, CalEx N=9, dnCx43 N=16. **(G)** Representative c-Fos staining in the lateral amygdala in control (eGFP), CalEx, and dnCx43 conditions. Scale bars = $10\mu\text{m}$. **(H)** c-Fos positive cell density was increased with astrocyte dysfunction. eGFP vs CalEx: $P<0.0001$; eGFP vs dnCx43: $P=0.0005$; CalEx vs dnCx43: $P=0.88$. eGFP n=17 slices N=5 mice, CalEx n=17 slices N=5 mice, dnCx43 n=12 slices N=5 mice. See Table S1 for detailed statistical summaries.

conditions, compared to viral control (**Figure 5F** [↗](#)). This effect was dependent on the increase in freezing to the neutral cue with freezing responses to the aversive tone being unaltered by astrocytic dysfunction (**Figure 5 – figure supplement 4B** [↗](#)). Taken together, these data directly implicate astrocytes in valence processing in the lateral amygdala and suggest a potential link between ELS and astrocyte dysfunction.

Investigation of potential sex differences in these genetic manipulations of astrocyte function revealed that the impact of CalEx on discrimination memory was more robust in female mice compared to males (**Figure 5 – figure supplement 4C** [↗](#)). In the case of dnCx43, no differences were observed between females and males (**Figure 5 – figure supplement 4D** [↗](#)).

Next, we set out to determine whether astrocyte dysfunction also mirrored the cellular and synaptic changes observed in ELS. Using acute brain slice electrophysiology, we report an occlusion of LTP in both CalEx and dnCx43 conditions (**Figure 5G** [↗](#)). In line with fear memory and synaptic plasticity changes, we also report that astrocyte dysfunction directly influences neural excitability with increased c-Fos density following fear conditioning in both CalEx and dnCx43 compared to control virus (**Figure 5H, I** [↗](#)). No sex differences were observed for c-Fos densities in eGFP- (**Figure 5 – figure supplement 5A, D** [↗](#)) or CalEx-expressing mice (**Figure 5 – figure supplement 5B, E** [↗](#)). dnCx43 expression however, resulted in higher c-Fos density in female mice compared to male counterparts (**Figure 5 – figure supplement 5F** [↗](#)).

Globally, these data highlight astrocytes as critical regulators of lateral amygdala function and output with astrocyte integrity essential for regulating synaptic plasticity, neural excitability, and of brain region-dependent behaviour.

Discussion

In this study, we investigated the potential roles of astrocytes in cellular and behavioural dysfunctions associated with ELS. Using a rodent model of ELS that employs maternal separation and limited bedding, we report increased blood glucocorticoid levels associated with heightened anxiety-like behaviours. These endocrine and behavioural changes were accompanied by impaired discriminative fear memory in an amygdala-dependent auditory discrimination task. More precisely, we observed that ELS mice displayed increased fear responses to neutral cues with no change in freezing responses to conditioned stimuli. These behavioural changes were further associated with alterations on the cellular scale including synaptic plasticity and neuronal excitability. Investigating contributions of non-neuronal cells, we found robust changes in lateral amygdala astrocytes following ELS comprising increased glucocorticoid receptor translocation to the nucleus and decreased expression of key astrocyte network proteins. Functionally, we observed astrocyte hypoactivity with decreased frequency, amplitude, and size of calcium events following ELS.

To link astrocyte dysfunction with cellular, synaptic, and behavioural changes, we genetically targeted lateral amygdala astrocytes to mimic the effects we found in ELS. Astrocyte dysfunction alone, either decreasing gap junction coupling or reducing astrocyte calcium activity, phenocopied the effects of ELS across excitability, synaptic, and behavioural scales. Astrocytes have long been shown to be critical for memory formation, with the vast majority of studies reporting that astrocyte dysfunction impairs memory across various brain regions ([32](#) [↗](#), [34](#) [↗](#), [62](#) [↗](#), [63](#) [↗](#)). Here, we show that astrocyte dysfunction in the lateral amygdala effectively impairs discriminative fear memory evidenced by fear generalisation, *i.e.* the inappropriate expression of fear towards stimuli of neutral valence. These data reveal that astrocytes are critical for the appropriate behavioural responses to emotionally salient cues and provide evidence supporting the recently proposed contextual guidance theory of astrocyte function in the brain ([64](#) [↗](#)).

Using an ELS paradigm that coincides with peak astrocyte postnatal development (65,66), we found a latent increase in circulating glucocorticoids after ELS with peak levels appearing during late adolescence and subsiding towards adulthood. While the lack of endocrine response to stress during the stress paradigm might appear paradoxical, this ELS paradigm overlaps with the stress hyporesponsive period, a time window during which stressors fail to elicit increases in blood corticosterone levels (67–70). Indeed, similar to our findings, others have reported latent increases in blood corticosterone levels following early-life stress that were associated with increased anxiety-like behaviour (47) and fear generalisation (71) later in life. Increased levels of blood glucocorticoids had a direct impact on astrocytes with increased translocation of GR from the cytosol to the nucleus of astrocytes following ELS, indicative of a higher levels of glucocorticoid signaling (56). While the effects of ELS on blood glucocorticoids were transient, this could have long-term consequences on protein expression levels resulting in changes in stress susceptibility and behavioural deficits. The primary behavioural phenotype revealed by this work, i.e. fear generalisation, is a common trait observed in anxiety disorders (49,72,73). As such, these findings provide further insight into the neurobiological mechanisms of stress on endocrine, cellular, and behavioural scales.

Our work builds upon a growing literature that is interested in the potential roles of astrocytes in early-life stress (74–77). Here, we identified specific astrocytic impairments induced by ELS and followed up with precise genetic manipulation of these astrocyte functions in the lateral amygdala. These astrocyte-specific manipulations replicate cellular and fear generalisation phenotype observed following ELS. We did not, however, observe an anxiety-like phenotype using elevated plus maze and open field with amygdala astrocyte manipulation which could be interpreted as a lack of astrocytic involvement in these discrete behaviours.

We found that both ELS and astrocyte dysfunction enhance neuronal excitability, assessed by local c-Fos staining in the lateral amygdala following auditory discriminative fear conditioning. One interpretation of these data is that astrocytes might tune engram formation, with astrocyte dysfunction, genetically or after stress, increasing c-Fos expression and a loss of specificity of the memory trace. In support of this notion, evidence of a cellular mechanism whereby c-Fos levels inversely correlate with memory accuracy have been recently shown in hippocampal circuits (48). In addition, early life experience strongly shapes episodic memory development, with stress delaying maturation of memory processes (78). Further support comes from recent demonstration of astrocyte-engram neuron interactions with astrocytes potentially forming part of engram stabilisation or even direct recruitment of astrocytes into engrams (55,79). Our data are in line with previous reports that have demonstrated clear roles for astrocytes in memory formation and consolidation (33,37,38) and reveal that astrocytes participate in the neurobiological mechanisms by which stress influences emotional memory.

Our work revealed unexpected sex differences in calcium signalling in lateral amygdala astrocytes. We found astrocyte calcium events were larger in amplitude and size in female mice compared to males. This higher basal activity could explain why CalEx induced a more profound impairment in fear discrimination in female mice. Interestingly, these sex differences are eliminated by stress inducing a homogenous calcium signature between both sexes. Sex differences in astrocytic development and function is starting to gain increased attention (62,65,80), with much more to be learned regarding the potentially divergent functions of astrocytes in the female and male brain.

In sum, our data shed new light on the role of astrocytes as central regulators of amygdala-dependent behaviour, synaptic function, and excitability. In addition, our work suggests a key role of astrocytes in mediating stress-induced behavioural and cellular impairments, supporting the notion that astrocytic disruption is not simply downstream of neuronal dysfunction. These ideas are forcing a change in perception of how brain disorders develop, moving away from a neuro-centric cascade and placing astrocytes in the spotlight (64,81–85). Collectively, these data

support the hypothesis that astrocytes are key to execute the appropriate behavioural response to emotionally salient cues and underscore the potential of astrocytes as therapeutic targets in stress-related psychiatric conditions.

Methods

Animals

All experiments were performed in accordance with the guidelines for maintenance and care of animals of the Canadian Council on Animal Care (CCAC) and approved by the Institutional Committee for the Protection of Animals (CIPA) at the Centre Hospitalier de l'Université de Montréal. Both male and female C57BL/6J mice (Jax #000664) were used in the study, housed on a 12hr:12hr light:dark cycle (lights on at 06:30) with *ad libitum* access to food and water.

Early Life Stress protocol

ELS was performed as previously described (6). C57BL/6J pups were separated from their mothers for 4-hours per day (ZT2-ZT6) for 7 days between ages P10 and P17. During separation pups and mothers were placed into new cages with 60% less bedding. Bedding was weighed on P10 and divided 33% into each cage (home cage, pup separation cage, and mother separation cage). Separated pups and mothers had *ad libitum* access to food and water and pup cages were placed on a water heating pad maintained at 34 degrees Celsius. After the final day of separation (P17) bedding was returned to the original amount and pups were housed with their mothers until weaning at P21.

Behavioural assays

Auditory Discriminative Fear Conditioning

Mice between ages P45-70 were used. Four days prior to behavioural testing mice were single housed in individual cages. Two days later mice were handled for approximately 10 minutes by hand and mobilisation tube. On the first day of ADFC, single housed mice were left for 1 hour in behavioural room to habituate prior to the start of the conditioning assay.

Conditioning: Mice were first exposed to a neutral stimulus (CS-; white noise, 20s) repeated six times. After a two-minute interval mice were exposed to a tone which becomes the conditioned stimulus (CS+; 12 kHz tone, 20s) as it co-terminates with a mild foot shock (Unconditioned stimulus (US); 0.5mA, 2s) delivered in the last two seconds of tone presentation (repeated six times). Conditioning was carried out in chamber with checkered walls that was cleaned with 70% ethanol between animals.

Memory recall: 24 hours after conditioning mice were exposed to CS- and CS+ (counterbalanced) in a novel environment characterised by stripped walls and a white floor covering, cleaned with 0.5% hydrogen peroxide between animals. Fear learning was quantified by the percentage of time spent freezing to CS- and CS+/US during the conditioning phase. Fear memory was quantified by the percentage time freezing to CS- or CS+ following presentation in the memory-recall phase. A discrimination index was calculated comparing fear responses to CS+ and CS- using the following formula;

$$\text{Discrimination Index (DI)} = \frac{(\text{CS}^+ \text{Freezing}) - (\text{CS}^- \text{Freezing})}{(\text{CS}^+ \text{Freezing}) + (\text{CS}^- \text{Freezing})}$$

Open Field Task

This task comprised of a total of 5 minutes and during that time the mice were left to freely explore the OFT apparatus, a white plexiglass box measuring 38×38 cm. During the task, mice were filmed, and the time spent in distinct parts of the box *i.e.* centre and periphery were quantified using AnyMaze software (Stoelting). The total time spent in the center was used as a quantification of anxiety-like behaviour, where more time in the center represented less anxiety-like behaviour and the total distance travelled was measured to quantify locomotor changes.

Elevated Plus Maze

This task comprised of a total of 5 minutes during which mice were removed from their cages and placed at the centre of the four arms of the maze, facing an open arm. Mice were video recorded, and these videos were analysed using Anymaze software (Stoelting) which quantified the time spent (s) in closed arms or open arms during the 5min testing period. Anxiety-like behaviour was quantified by ratio of time spent in open vs closed arms.

Acute brain slice preparation

Coronal slices containing lateral amygdala were obtained from mice between P45-70. Animals were anesthetized with isoflurane and the brain was rapidly excised and placed in ice-cold choline-based cutting solution saturated with 95% O₂ and 5% CO₂ containing the following (in mM): 120 choline chloride, 3 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 8 MgCl₂, 20 glucose, pH 7.2 and 295 mOsmol. Slices (300 µm thick) were cut on a vibratome (Leica VT1200, Nussloch, Germany). Slices were transferred to oxygenated artificial CSF (ACSF) at 32 ± 0.5°C for 30 min containing the following (in mM): 130 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 10 glucose, 1.3 MgCl₂, 2 CaCl₂, pH 7.3–7.4 and 300–305 mOsmol, then allowed to recover for at least 30 min before recordings in oxygenated ACSF at room temperature (RT). For experiments, the slices were transferred to a recording chamber where they were perfused (2 ml/min) with ACSF at 32°C for the course of the experiment. Slices were used for a maximum of 6 hr after cutting.

Electrophysiology

Extracellular field excitatory postsynaptic potentials (fEPSPs) were recorded in current clamp mode with a pipette filled with ACSF placed in the lateral part of the amygdala. Monosynaptic EPSPs were elicited by stimulation of cortical afferences with a tungsten concentric bipolar electrode (World Precision Instruments). LTP was induced using a theta burst stimulation (TBS) protocol consisting of 4 pulses at 100 Hz repeated at 5 Hz intervals for 10 seconds. Signals were amplified using a Multiclamp 700B amplifier (Molecular Devices) and digitized using the Digidata 1440 (Molecular Devices). Data were recorded (pClamp 10.2, Molecular Devices) for offline analysis. The magnitude of LTP was quantified by normalizing and averaging the slope of fEPSPs following LTP induction relative to the last 5 min of baseline.

Two-photon calcium imaging and analysis

Fluorescence calcium imaging was performed on a custom two-photon laser-scanning microscope (Scientifica). The microscope was equipped with a Ti:Sapph laser (Ultra II, Coherent), a green bandpass emission filter (525–40 nm), an orange/red bandpass emission filter (605–70 nm), and associated photomultiplier tubes (GaAsP, Hamamatsu). Time-series images were acquired through bidirectional laser scanning (920nm excitation laser, 86.85µm x 86.85µm, 512×512 pixels at 1Hz) at a single focal plane incorporating the entirety of the imaged astrocyte ramifications. All images were acquired using Scanimage (mbf bioscience). Acute brain slices (350µm) containing the lateral amygdala were obtained from mice (ages between P45-70) injected with the viral construct AAV2/5-gfaABC1D-lck-GCaMP6f 3 weeks prior. Animals were anesthetized with isoflurane and the brain was rapidly excised and placed in ice-cold NMDG based cutting solution saturated with 95%

O₂ and 5% CO₂ containing the following (in mM): 119.9 NMDG, 2.5 KCl, 25, 1 CaCl₂, 1.4 NaH₂PO₄ and 20 D-glucose saturated with 95% O₂ and 5% CO₂. Slices (350µm thick) were cut on a vibratome (Leica VT1200, Nussloch, Germany). Slices were transferred to oxygenated NMDG cutting solution artificial CSF (aCSF) at 33 ± 0.5°C for 15 min followed by a 1-hour recovery period in artificial CSF (aCSF) at RT containing the following (in mM): 130 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 2.5 glucose, 1.3 MgCl₂, 2 CaCl₂, pH 7.3–7.4 and 300–305 mOsmol. All time-series images were acquired in a recording chamber constantly perfused with aCSF as described above.

For unbiased, automatic calcium event detection, TIFF files of time-lapse images were processed with the AQUA2 pipeline in MATLAB. Parameters of interest (event frequency, amplitude, area, duration, rise time and decay time) were directly extracted from the AQUA2 output file. For statistical testing, averages of all events from a single astrocyte were calculated for each parameter. In parallel, to better represent the diversity in calcium events, frequency distributions plots were created considering every detected event for all our experimental conditions.

Stereotaxic surgery and AAV delivery

P25–35 mice were given a subcutaneous (s.c.) injection of Carprofen (20 mg/kg) 1h prior surgery. Animals were then deeply anesthetised (2% isoflurane) before placing them into a stereotaxic frame (Kopf instruments). Surgical site was infiltrated with bupivacaine/lidocaine (2mg/kg sc.) 15 min prior surgery. Bilateral injections of the following viral constructs produced at the Canadian Neurophotonics Viral Vector Core: AAV2/5-gfaABC1D-lck-GCamp6f (9.6×10^{12} GC/ml), AAV2/5-gfaABC1D-eGFP ($1.2\text{--}1.5 \times 10^{11\text{--}12}$ GC/ml), AAV2/5-gfaABC1D-dnCx43-eGFP ($1.2\text{--}1.7 \times 10^{12}$ GC/ml) and AAV2/5-gfaABC1D-hPMCAw/b-mCherry (1.2×10^{12}) were delivered to the LA (coordinates relative to bregma: −1.4 mm AP, ± 3.4 mm ML and −5 mm DV) of mice of either sex. Viral solutions were injected at a rate of 200 nL/min for a total volume of 650 nL per hemisphere using a 10 µl Hamilton syringe (Harvard apparatus) and a microinjection syringe pump (UMP3T-2 World Precision Instruments). Animals received 250 µL of saline for post-surgery hydration.

Brain processing

Mice were anaesthetised in an enclosed chamber filled with isoflurane (5% for induction, 3–4% for maintenance, v/v) until loss of toe pinch reflex. A transcardial perfusion was used to ensure uniform preservation of tissue and brains were fixed in 4 % paraformaldehyde (PFA) for 24 hours at 4 degrees Celsius. The fixed brains were then placed in falcon tubes containing 30% sucrose solution for a minimum of two days. Brains were embedded in optimal cutting temperature (OCT) compound and flash frozen in 2-methylbutane between −40 and −50 degrees Celsius and then stored at −80 until cryosectioning. Brains were cut at 30µm thickness at −20 degrees Celsius using a Leica CM3050S Cryostat. Slices were stored in cryoprotector solution at 4 degrees Celsius until immunohistochemistry.

For c-fos labeling experiments, mice were anaesthetised and then perfused 90 minutes after the end of fear conditioning. The whole lateral amygdala was sectioned in 30µm thick brain slices and every fourth brain slice was selected for immunohistochemistry.

Immunohistochemistry

Free floating brain slices were washed 3 times in 1X PBS for 15 minutes to remove cryoprotector solution. Slices were permeabilised in block-perm solution (3% Bovine-Serum-Albumin, 0.5% Triton^{10%} in PBS) for a duration of 1 hour. Slices were incubated with antibodies for 24 hours at 4 degrees Celsius. Slices were incubated with primary antibodies; [Rabbit] anti-S100β (1:1000, Abcam, ab52642), [Mouse] anti-glucocorticoid receptor (1:500, ThermoFisher, MA1-510), [Chicken] anti-GFAP (1:1000, ThermoFisher, PA1-10004), [mouse] anti-Cx43 (1:1000 ThermoFisher, 3D8A5), [Rabbit] Anti-GLT-1 (EAAT2) (1:1000, NovusBio, NBP1-20136) and), [Rabbit] Anti-c-Fos (1:2000, Synaptic Systems, Cat.No. 226008). After primary antibody incubation slices were washed 3 times

in 1X PBS for 15 minutes to remove non-specific binding. Slices were incubated with secondary antibodies in a DAPI solution (1:10000) at RT in aluminium foil for 1 hour. Slices were incubated with secondary antibodies; [goat] anti-rabbit Alexa 488 (1:1000, Jackson Immuno Research, 111-545-144), [goat] anti-mouse Alexa 647 (1:1000, ThermoFisher, A32728), [goat] anti-chicken Alexa 568 (1:1000, ThermoFisher, A11041). After secondary antibody incubation slices were washed again 3 times in 1X PBS for 15 minutes before being mounted onto Fisherbrand microscope slides using ProLong™ Glass Antifade mountant (P36982).

Microscopy

Slices were imaged using a Leica TCS SP5 laser scanning confocal microscope with oil immersion Plan-Apochromat 63x objective 1.4 NA, or using a Zeiss Observer Z1 spinning disk confocal microscope/TIRF with a 20x objective 1.8 NA. 16-bit images of 246 x 246 μm areas were acquired at 400Hz (frame size (x*y); 1024 x 1024, pixel size; 250nm). 25-30 μm z-stacks were acquired with a step size of 0.5 μm . 15 μm max intensity z-projections of the lateral amygdala were analysed using Image-J (Fiji) to obtain A.U fluorescence intensity measures of secondary antibodies attached to primaries with specificity to epitopes; GFAP, Cx43, GLT-1, and GR. We applied a fluorescence threshold for GFAP, Cx43 and GLT-1 fluorescence and measured integrated density A.U (thresholds; Mean, Otsu, Otsu respectively). For nuclear and cytosolic Glucocorticoid Receptor measures, astrocyte (S100 β^+ DAPI nuclei were thresholded and used as ROIs for nuclear measures of GR fluorescence (S100 β^+ + DAPI $^+$). The same nuclear DAPI ROI was cleared from the previously thresholded s100b regions to obtain cytosolic astrocytic GR (S100 β^+ - DAPI $^+$). Integrated density A.U of GR fluorescence was used to calculate Nuclear:Cytosolic GR ratios.

For c-Fos cell counting in the lateral amygdala, 16-bit images containing the lateral and basal amygdala nuclei were acquired. 20-30 z-stacks were acquired with a step size of 1 μm . For c-Fos particle counting analysis, a region of interest was used to isolate the lateral amygdala and then 8-bit z-stacks of the c-Fos channel were analysed semi-automatically by using the FIJI plugin Quanty-cfos (86 [86](#)). In brief, a threshold for area and fluorescence intensity was set manually to identify ROIs around all c-Fos positive nuclei. These ROIs were then manually inspected to remove false positives and negatives found by the plug-in. c-Fos particle counts were then normalized by the volume of the z-stack to calculate the c-Fos density. In order to calculate the volume, the area of the lateral amygdala was multiplied by the number of z-stacks.

Blood Collection & Corticosterone analysis

Trunk blood was collected via decapitation at ZT2 (08:30). To minimise handling induced elevations in corticosterone mice were housed with clear plastic tubes used to move individual mice into an enclosed chamber filled with 5% isoflurane for 2 minutes. Mice were placed in enclosed Isoflurane until loss of toe pinch reflect (maximum elapsed time of 2 minutes). Mice were decapitated and trunk blood was collected into BD 365963 Microtainer® Capillary Blood Collection Tubes and placed onto Ice. Blood was centrifuged for 5 minutes at 4 degrees Celsius at 5000rpm. Serum was aliquoted and stored at -80 degrees Celsius. Corticosterone measurements were obtained using an ENZO ELISA kit (ADI-900-097).

Statistical analyses

Results are presented as mean $\pm$ S.E.M. Data with one variable were analyzed with the two-tailed Student's *t* test or Mann-Whitney test. Data with more than two conditions were first screened for a Gaussian distribution with Kolmogorov-Smirnov test followed by analysis either with one-way/repeated measures ANOVA or Kruskal-Wallis/Friedman test when needed and Tukey's multiple-comparison parametric *post hoc* test (data with Gaussian distribution) or by a Dunn's multiple-comparison non-parametric *post hoc* test (data with non-Gaussian distribution). Auditory discriminative fear conditioning data was analysed using a two-way ANOVA with appropriate

post-hoc tests. Open field and elevated plus maze data analysed using one-way ANOVA with appropriate post hoc tests. Graphic significance levels were *, $p < 0.05$; **, $p < 0.01$ and ***, $p < 0.001$. All data were analyzed using GraphPad Prism software (Version 9, GraphPad, USA).

Data availability

The datasets that support the findings of this study are available from the corresponding author upon request.

Supplementary figures and tables

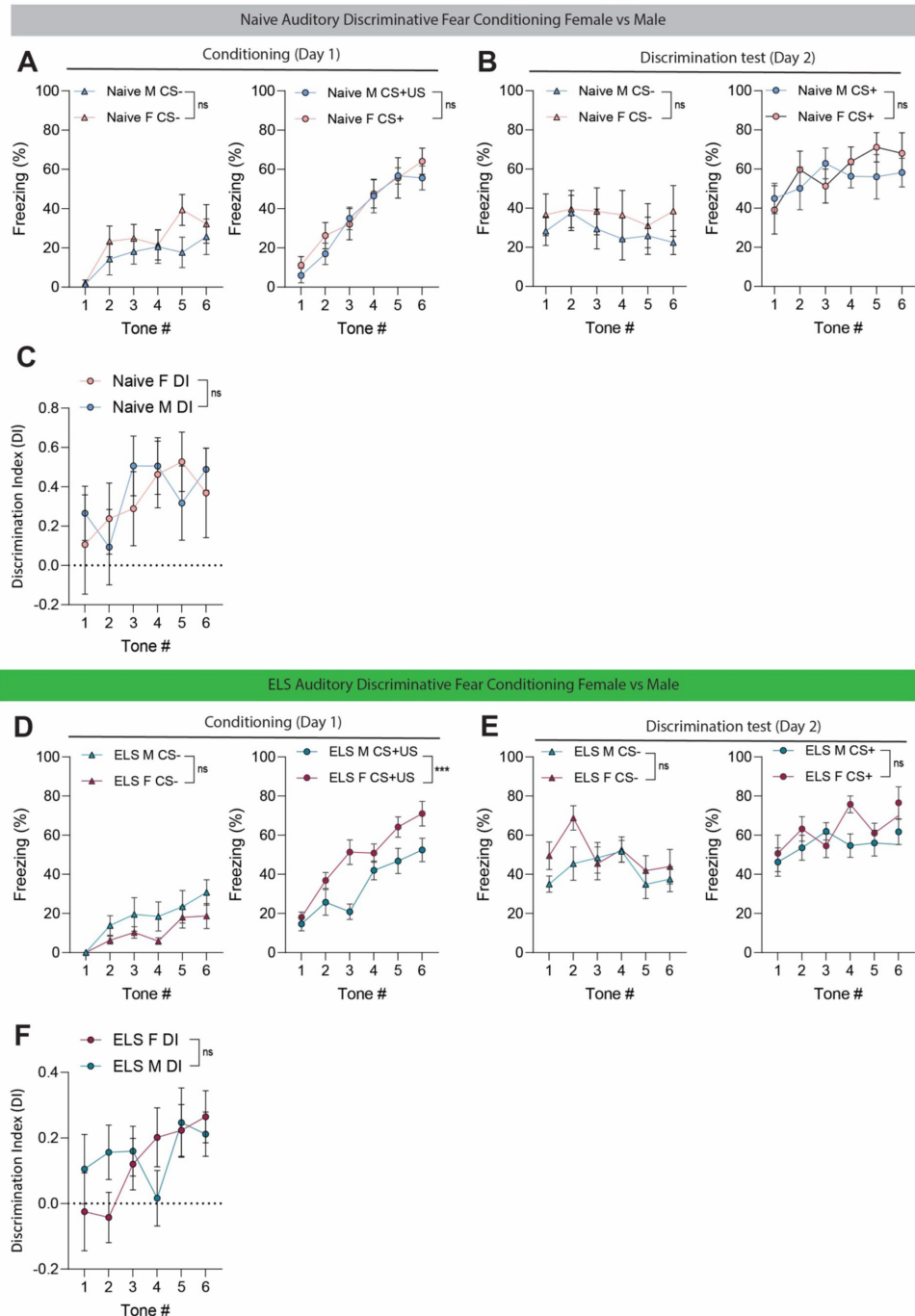


Figure 2 – Figure supplement 1

(A) No differences between male and female naïve mice freezing during learning phase (conditioning) of auditory discriminative fear conditioning to CS- (left), $P=0.2521$; or CS+ (right), $P=0.8432$. Female $N=7$, Male $N=8$. **(B)** No differences between male and female naïve mice during memory recall of CS- (left), $P=0.3956$; and CS+ (right), $P=0.8851$. Female $N=7$, Male $N=8$. **(C)** No differences in auditory discrimination between male and female naïve mice. $P=0.8505$, Female $N=7$, Male $N=8$. **(D)** No differences between male and female ELS mice freezing during learning phase (conditioning) of auditory discriminative fear conditioning to CS- (left), $P=0.2521$. Female ELS mice freeze significantly more to CS+ (right) than male counterparts, $P<0.0001$. Female $N=9$, Male $N=8$. **(E)** No differences between male and female ELS mice during memory recall of CS- (left), $P=0.16220$; or CS+ (right), $P=0.1869$. Female $N=9$, Male $N=8$. **(F)** No differences in auditory discrimination between male and female ELS mice. $P=0.5157$, Female $N=9$, Male $N=8$.

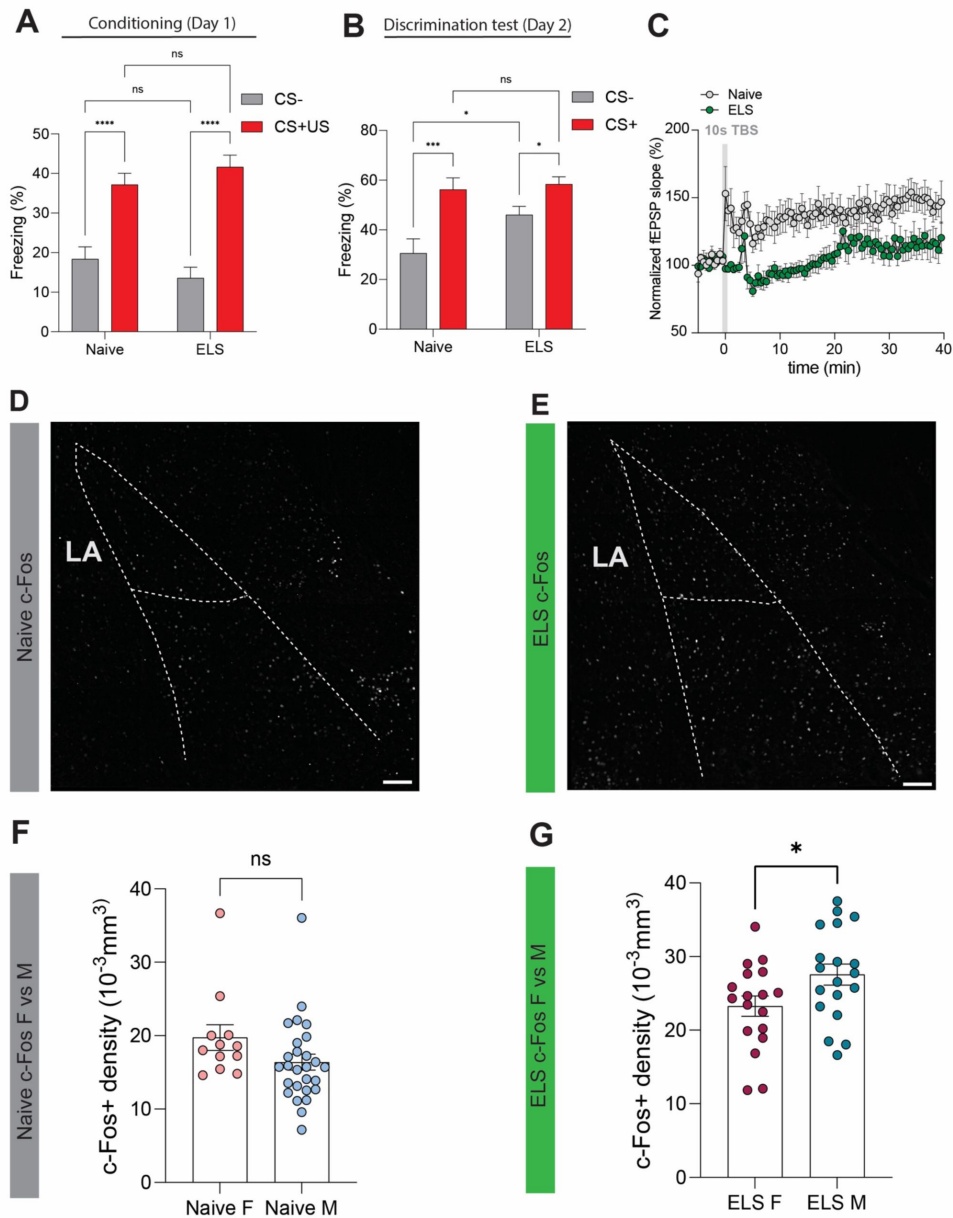


Figure 2 – Figure supplement 2

(A) Average freezing responses to CS- (grey) and CS+ (red) during fear conditioning for naïve and ELS mice. Tone (CS +/-) represents the within-subject variable while treatment (naïve/ELS) represents the between-subject variable. **Naïve (CS- vs CS+/US)**, $P < 0.0001$, $N = 15$ mice; **ELS (CS- vs CS+/US)**, $P < 0.0001$, $N = 17$; **CS- (Naïve vs ELS)**, $P = 0.2543$, Naïve $N = 15$ mice, ELS $N = 17$ mice; **CS+/US (Naïve vs ELS)**, $P = 0.2851$, Naïve $N = 15$ mice, ELS $N = 17$ mice. **(B)** Average freezing responses to CS- (grey) and CS+ (red) during Discrimination for naïve and ELS mice. Tone (CS +/-) represents the within-subject variable while treatment (naïve/ELS) represents the between-subject variable. **Naïve (CS- vs CS+/US)**, $P = 0.0001$, $N = 15$ mice; **ELS (CS- vs CS+/US)**, $P = 0.0392$, $N = 17$ mice; **CS- (Naïve vs ELS)**, $P = 0.0128$, Naïve $N = 15$ mice, ELS $N = 17$ mice; **CS+/US (Naïve vs ELS)**, $P = 0.7267$, Naïve $N = 15$ mice, ELS $N = 17$ mice. **(C)** Normalised fEPSP slope showing synaptic response before and after theta burst stimulation (grey vertical bar) in naïve and ELS mice. **(D)** Representative Z-stack projection of c-Fos staining in entire lateral and basolateral amygdala in a naïve mice 90 minutes after conditioning. **(E)** Representative Z-stack projection of c-Fos staining in entire lateral and basolateral amygdala in an ELS mice 90 minutes after conditioning. **(F)** No differences in c-Fos density between male and female naïve mice 90 minutes after conditioning. $P = 0.1020$, Male $n = 27$, Female $n = 12$. **(G)** ELS results in increased c-Fos density 90 minutes after conditioning in male mice compared to female counterparts. $P = 0.0383$, Male $n = 19$, Female $n = 18$. Scale bars = $100\mu\text{m}$

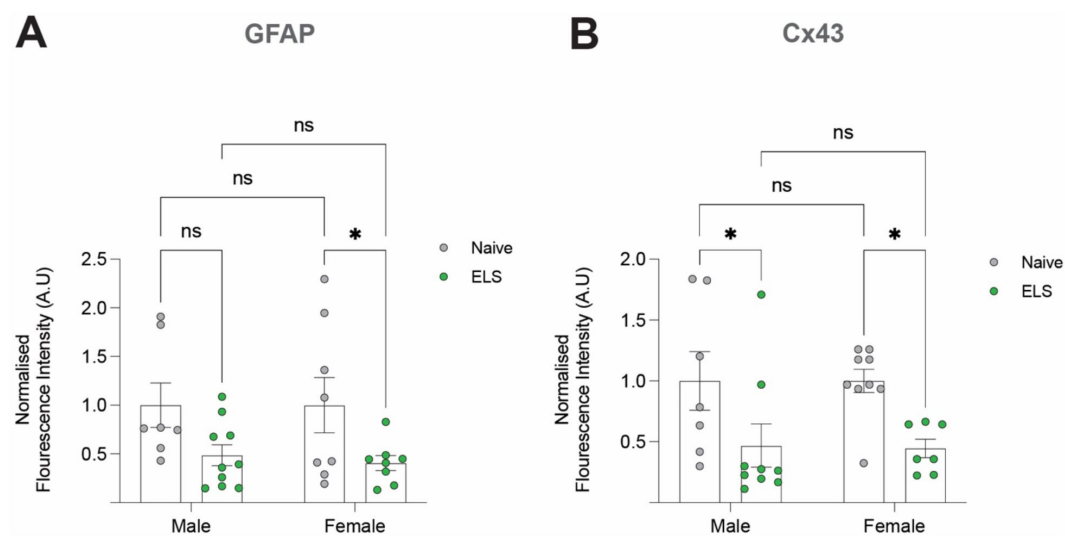


Figure 3 - Figure supplement 1

(A) Normalised Fluorescence intensity values of GFAP in the lateral amygdala of Naïve and ELS mice split by Sex. Naïve Male (N=7) vs ELS male (N=10), $P=0.582$; **Naïve Female** (N= 8) vs **ELS Female** (N=8), $P=0.0323$; **Naïve Male** vs **Naïve Female**, $P>0.999$; **ELS male** vs **ELS Female**, $P=0.750$. Normalised Fluorescence intensity values of Cx43 in the lateral amygdala of Naïve and ELS mice split by Sex. Naïve Male (N=7) vs ELS male (N=9), $P=0.0253$; **Naïve Female** (N= 9) vs **ELS Female** (N=7), $P=0.0201$; **Naïve Male** vs **Naïve Female**, $P>0.999$; **ELS male** vs **ELS Female**, $P=0.9189$.

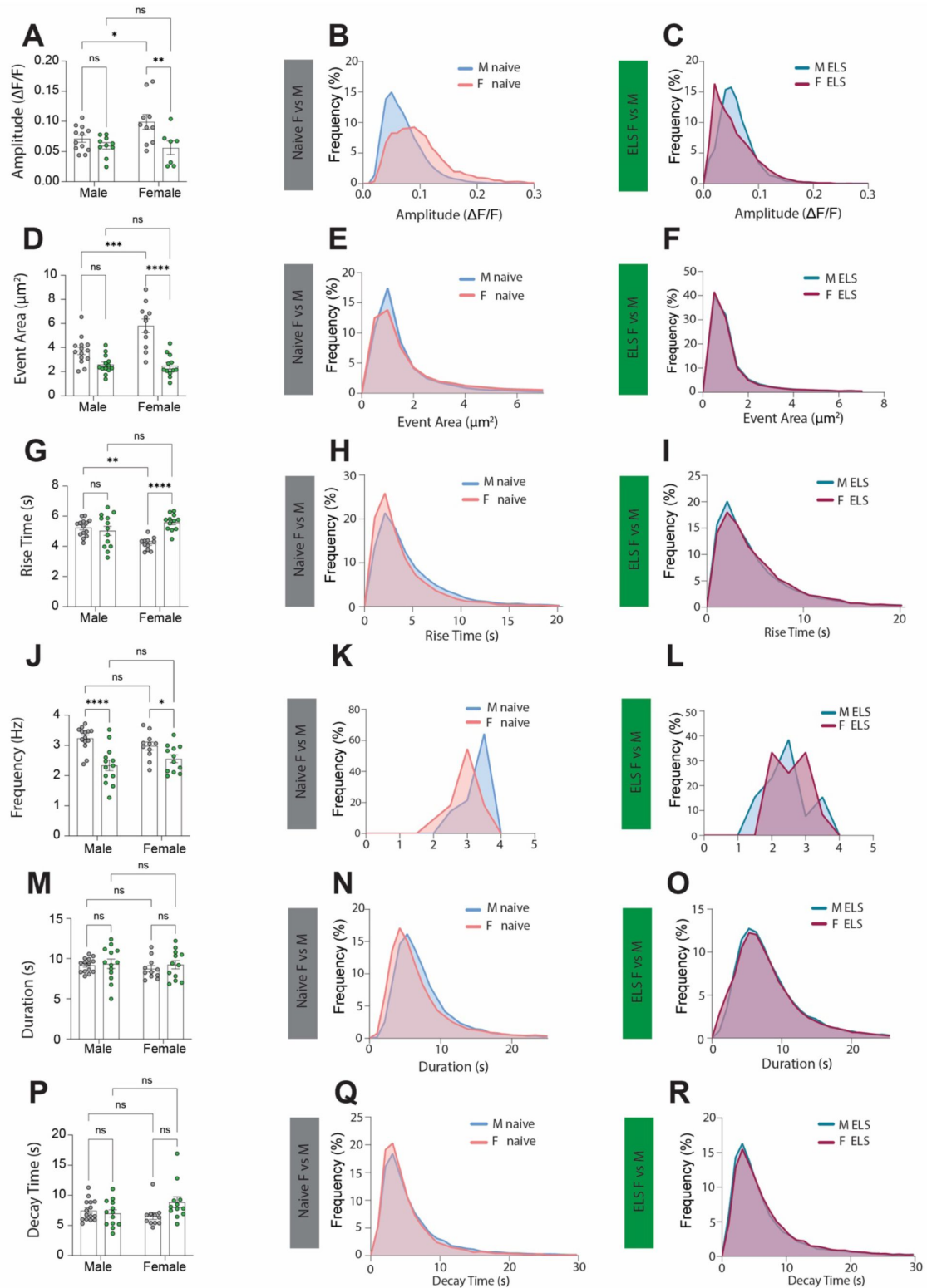


Figure 4 – Figure supplement 1

(A) ELS results in decreased calcium event amplitude only for female mice. **Male** (Naïve vs ELS), $P=0.5478$; **Female** (Naïve vs ELS), $P=0.0058$. Increased calcium event amplitude was observed in naïve females compared to males. **Naïve** (Male vs Female), $P=0.0425$. No differences were observed between male and female ELS animals. **ELS** (Male vs Female), $P=0.9627$. Naïve Male $n=13$ astrocytes, Naïve Female $n=10$ astrocytes, ELS Male $n=10$ astrocytes, ELS Female $n=8$ astrocytes. (B) Frequency distribution of calcium event amplitude from naïve male (blue line) and female (red line) mice. (C) Frequency distribution of calcium event amplitude from ELS male (dark teal line) and female (purple line) mice. (D) ELS results in decreased calcium event area only for female mice. **Male** (Naïve vs ELS), $P=0.0385$; **Female** (Naïve vs ELS), $P<0.0001$. Increased calcium event area was observed in naïve females compared to males. **Naïve** (Male vs Female), $P=0.0002$. No differences were observed between male and female ELS animals. **ELS** (Male vs Female), $P=0.9702$. Naïve Male $n=14$ astrocytes, Naïve Female $n=11$ astrocytes, ELS Male $n=13$ astrocytes, ELS Female $n=12$ astrocytes. (E) Frequency distribution of calcium event area from naïve male (blue line) and female (red line) mice. (F) Frequency distribution of calcium event area from ELS male (dark teal line) and female (purple line) mice. (G) ELS results in increased calcium event rise time only for female mice. **Male** (Naïve vs ELS), $P=0.6814$; **Female** (Naïve vs ELS), $P<0.0001$. Increased calcium event amplitude was observed in naïve males compared to females. **Naïve** (Male vs Female), $P=0.0014$. No differences were observed between male and female ELS animals. **ELS** (Male vs Female), $P=0.1062$. Naïve Male $n=14$ astrocytes, Naïve Female $n=11$ astrocytes, ELS Male $n=13$ astrocytes, ELS Female $n=12$ astrocytes. (H) Frequency distribution of calcium event rise time from naïve male (blue line) and female (red line) mice. (I) Frequency distribution of calcium event rise time from ELS male (dark teal line) and female (purple line) mice. (J) ELS results in decreased frequency of calcium events both for males and females. **Male** (Naïve vs ELS), $P<0.0001$; **Female** (Naïve vs ELS), $P=0.0419$. No difference in calcium event frequency were observed between male and females within the naïve or ELS groups. **Naïve** (Male vs Female), $P=0.2053$; **ELS** (Male vs Female), $P=0.2671$. Naïve Male $n=14$ astrocytes, Naïve Female $n=11$ astrocytes, ELS Male $n=13$ astrocytes, ELS Female $n=12$ astrocytes. (K) Frequency distribution of calcium event frequency from naïve male (blue line) and female (red line) mice. (L) Frequency distribution of calcium event frequency from ELS male (dark teal line) and female (purple line) mice. (M) No differences in calcium event duration between Naïve and ELS female and male mice. **Male** (Naïve vs ELS), $P=0.7688$; **Female** (Naïve vs ELS), $P=0.4499$. **Naïve** (Male vs Female), $P=0.4983$; **ELS** (Male vs Female), $P=0.8604$. Naïve Male $n=14$ astrocytes, Naïve Female $n=11$ astrocytes, ELS Male $n=13$ astrocytes, ELS Female $n=12$ astrocytes. (N) Frequency distribution of calcium event duration from naïve male (blue line) and female (red line) mice. (O) Frequency distribution of calcium event duration from ELS male (dark teal line) and female (purple line) mice. (P) No differences in calcium event decay time between Naïve and ELS female and male mice. **Male** (Naïve vs ELS), $P=0.9997$; **Female** (Naïve vs ELS), $P=0.0718$. **Naïve** (Male vs Female), $P=0.5942$; **ELS** (Male vs Female), $P=0.3383$. Naïve Male $n=14$ astrocytes, Naïve Female $n=11$ astrocytes, ELS Male $n=13$ astrocytes, ELS Female $n=12$ astrocytes. (Q) Frequency distribution of calcium event decay time from naïve male (blue line) and female (red line) mice. (R) Frequency distribution of calcium event decay time from ELS male (dark teal line) and female (purple line) mice.

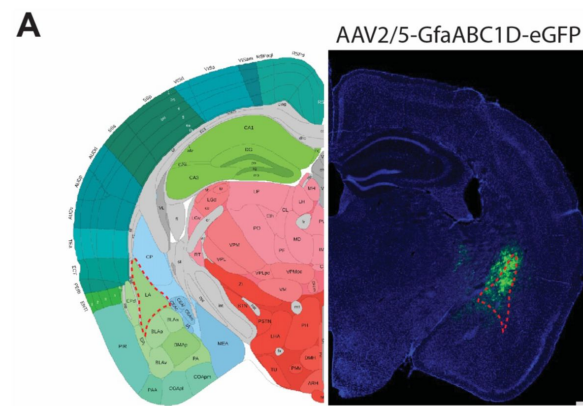


Figure 5 – Figure supplement 1

(A) Representative viral targeting of lateral amygdala astrocytes with AAV-GfaABC₁D-eGFP.

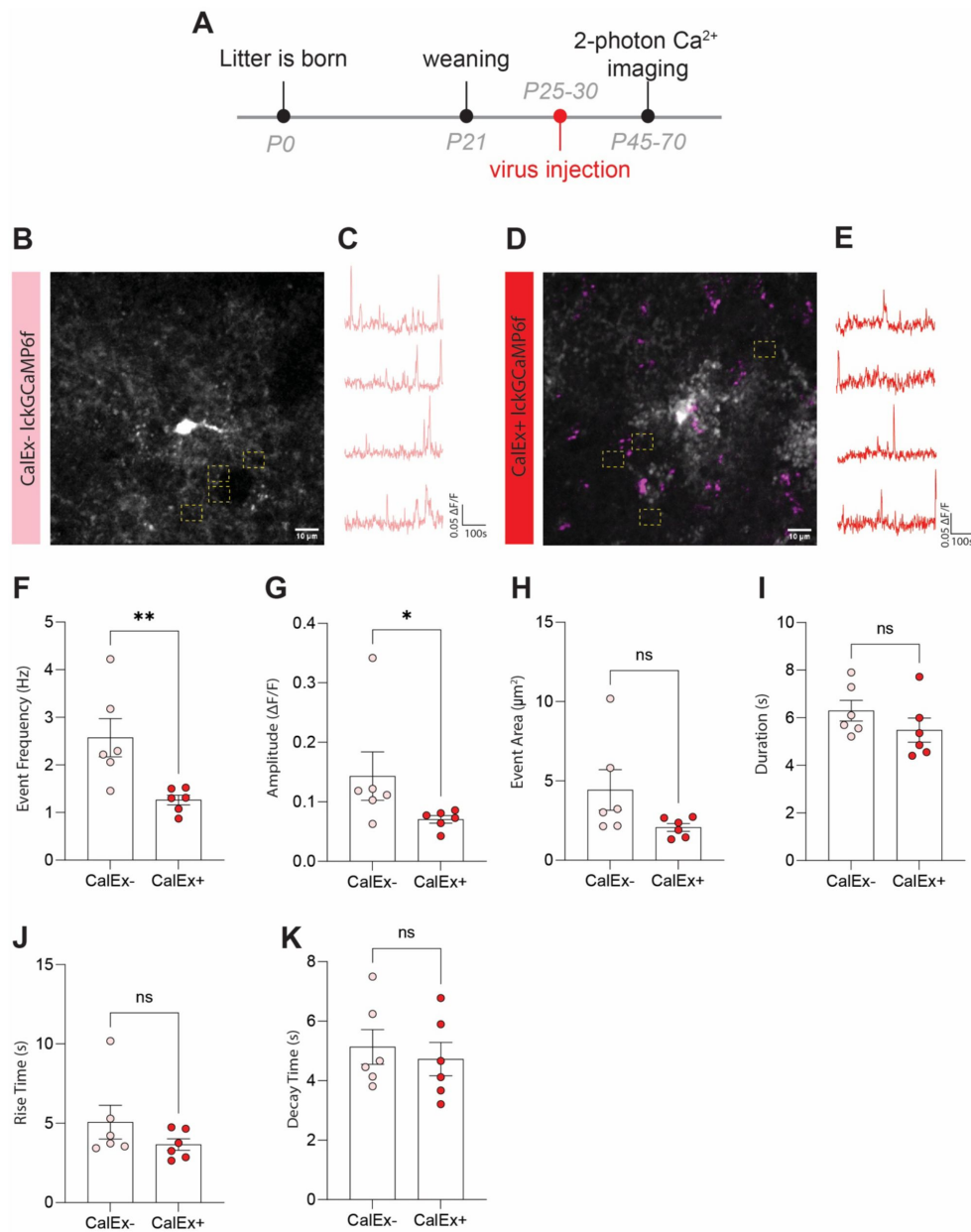


Figure 5 – Figure supplement 2

(A) Timeline of CalEx/lck-GCaMP6 viral injections and 2-photon calcium imaging experiments. **(B)** Representative image of a non-CalEx (CalEx-) lck-GCaMP6-expressing astrocyte in the lateral amygdala with 4 regions of interest from a naïve animal. Scale bar=10 μm **(C)** Representative calcium-signaling traces from 4 regions of interest shown in **B**. **(D)** Representative image of a CalEx+lck-GCaMP6-expressing astrocyte in the lateral amygdala with 4 regions of interest from an ELS animal. Scale bar=10 μm **(E)** Representative calcium-signaling traces from 4 regions of interest shown in **D**. **(F)** CalEx effectively decreases the frequency of calcium events in lateral amygdala astrocytes. $P=0.0087$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 animals. **(G)** CalEx decreases calcium event amplitude in lateral amygdala astrocytes. $P=0.0411$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 animals. **(H)** CalEx does not impact calcium event area in lateral amygdala astrocytes. $P=0.0649$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 animals. **(I)** CalEx does not impact calcium event duration in lateral amygdala astrocytes. $P=0.1797$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 animals. **(J)** CalEx does not impact calcium event rise time in lateral amygdala astrocytes. $P=0.3095$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 mice. **(K)** CalEx does not impact calcium event decay time in lateral amygdala astrocytes. $P=0.5887$, CalEx-n=6 astrocytes from N=3 animals, CalEx+ n=6 astrocytes from N=3 mice.

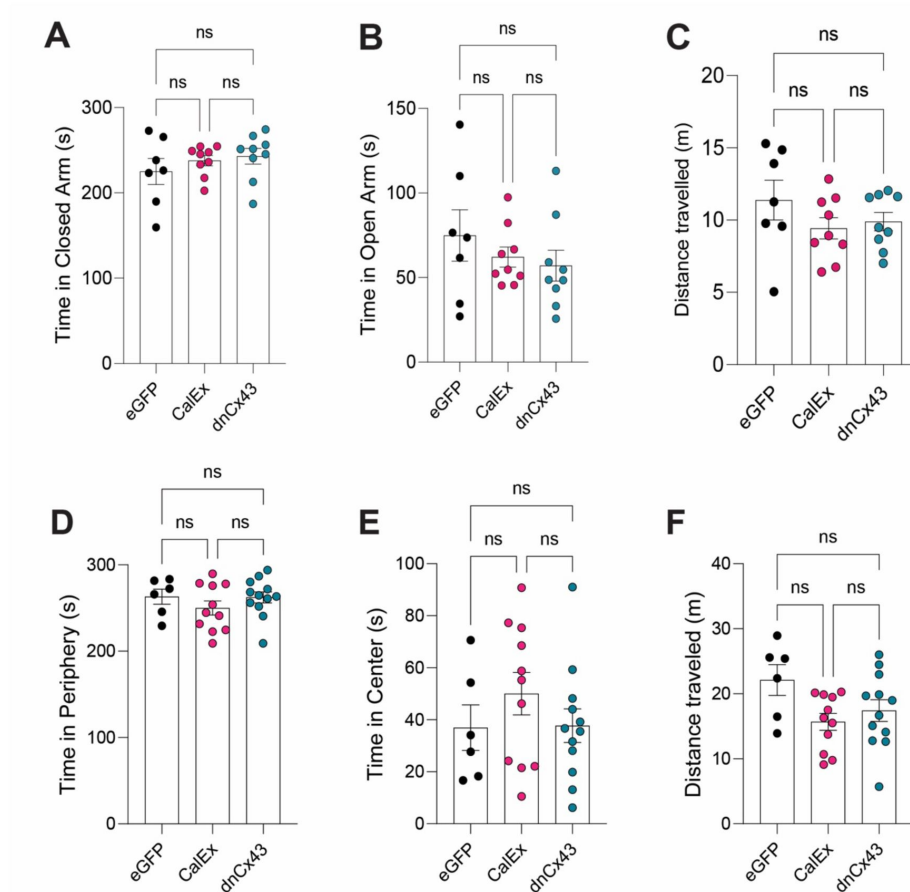


Figure 5 – Figure supplement 3

(A) No effect of lateral amygdala astrocyte viral manipulation on time spent in closed arms of elevated plus maze. **eGFP vs CalEx**; $P > 0.9999$; **eGFP vs dnCx43**, $P = 0.7177$; **CalEx vs dnCx43**, $P > 0.9999$. eGFP N= 7 animals, CalEx N=9 animals, dnCx43 N=9 animals. **(B)** No effect of lateral amygdala astrocyte viral manipulation on time spent in open arms of elevated plus maze. **eGFP vs CalEx**; $P > 0.9999$; **eGFP vs dnCx43**, $P = 0.7177$; **CalEx vs dnCx43**, $P > 0.9999$. eGFP N= 7 animals, CalEx N=9 animals, dnCx43 N=9 animals. **(C)** No effect of lateral amygdala astrocyte viral manipulation on total distance travelled in the elevated plus maze. **eGFP vs CalEx**; $P = 0.3095$; **eGFP vs dnCx43**, $P = 0.5025$; **CalEx vs dnCx43**, $P = 0.9197$. eGFP N= 7 animals, CalEx N=9 animals, dnCx43 N=9 animals. **(D)** No effect of lateral amygdala astrocyte viral manipulation on time spent in the periphery of open field maze. **eGFP vs CalEx**; $P = 0.8547$; **eGFP vs dnCx43**, $P > 0.9999$; **CalEx vs dnCx43**, $P = 0.9248$. eGFP N= 6 animals, CalEx N=11 animals, dnCx43 N=12 animals. **(E)** No effect of lateral amygdala astrocyte viral manipulation on time spent in the center of open field maze. **eGFP vs CalEx**; $P = 0.8547$; **eGFP vs dnCx43**, $P > 0.9999$; **CalEx vs dnCx43**, $P = 0.9248$. eGFP N= 6 animals, CalEx N=11 animals, dnCx43 N=12 animals. **(F)** No effect of lateral amygdala astrocyte viral manipulation on total distance travelled in the open field maze. **eGFP vs CalEx**; $P = 0.8547$; **eGFP vs dnCx43**, $P > 0.9999$; **CalEx vs dnCx43**, $P = 0.9248$. eGFP N= 6 animals, CalEx N=11 animals, dnCx43 N=12 animals.

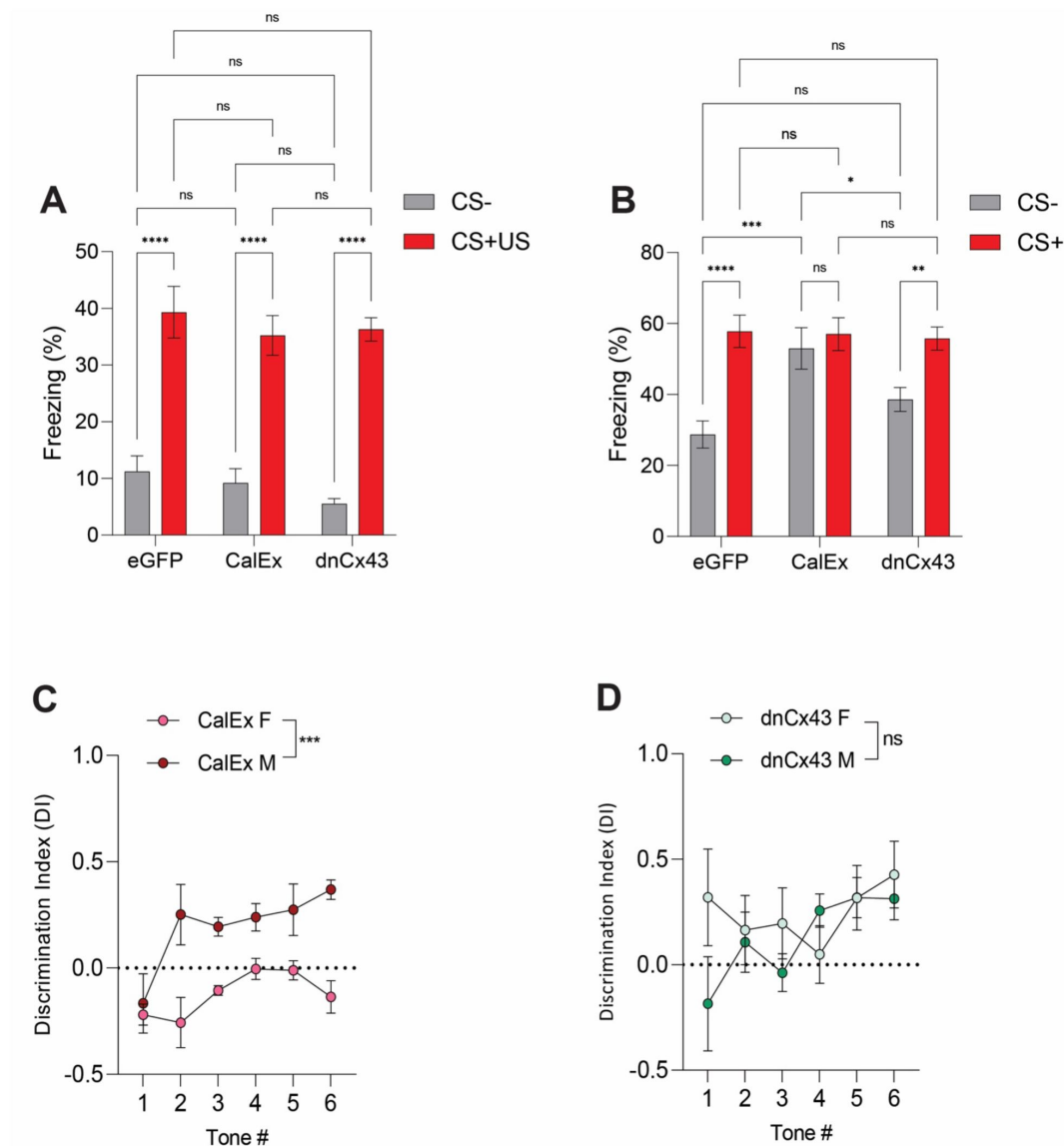


Figure 5 - Figure supplement 4

(A) Average freezing responses to CS- (grey) and CS+ (red) during fear conditioning for eGFP, CalEx and dnCx43 mice. Tone (CS +/-) represents the within-subject variable while treatment (naïve/ELS) represents the between-subject variable. **eGFP** (CS- vs CS+US), $P < 0.0001$, $N = 12$ animals; **CalEx** (CS- vs CS+US), $P < 0.0001$, $N = 11$ animals; **dnCx43** (CS- vs CS+US), $P < 0.0001$, $N = 17$ animals; **CS-** (eGFP vs CalEx), $P = 0.9448$; **CS-** (eGFP dnCx43), $P = 0.3372$; **CS-** (CalEx vs dnCx43), $P = 0.7121$; **CS+US** (eGFP vs CalEx), $P = 0.6879$; **CS+US** (eGFP vs dnCx43), $P = 0.8030$; **CS+US** (CalEx vs dnCx43), $P = 0.9891$. **(B)** Average freezing responses to CS- (grey) and CS+ (red) during Discrimination for eGFP, CalEx and dnCx43 mice. Tone (CS +/-) represents the within-subject variable while treatment (naïve/ELS) represents the between-subject variable. **eGFP** (CS- vs CS+US), $P < 0.0001$, $N = 12$ animals; **CalEx** (CS- vs CS+US), $P < 0.0001$, $N = 11$ animals; **dnCx43** (CS- vs CS+US), $P < 0.0001$, $N = 17$ animals; **CS-** (eGFP vs CalEx), $P = 0.9448$; **CS-** (eGFP dnCx43), $P = 0.3372$; **CS-** (CalEx vs dnCx43), $P = 0.7121$; **CS+US** (eGFP vs CalEx), $P = 0.6879$; **CS+US** (eGFP vs dnCx43), $P = 0.8030$; **CS+US** (CalEx vs dnCx43), $P = 0.9891$. **(C)** CalEx disruption of lateral amygdala astrocytes affects female auditory cue discrimination significantly more than male counterparts. $P < 0.0001$, Male $N = 6$ animals, Female $N = 5$ animals. **(D)** No differences in the impact of dnCx43 on auditory cue discrimination between male and female animals. $P = 0.1803$, Male $N = 10$, Female $N = 7$.

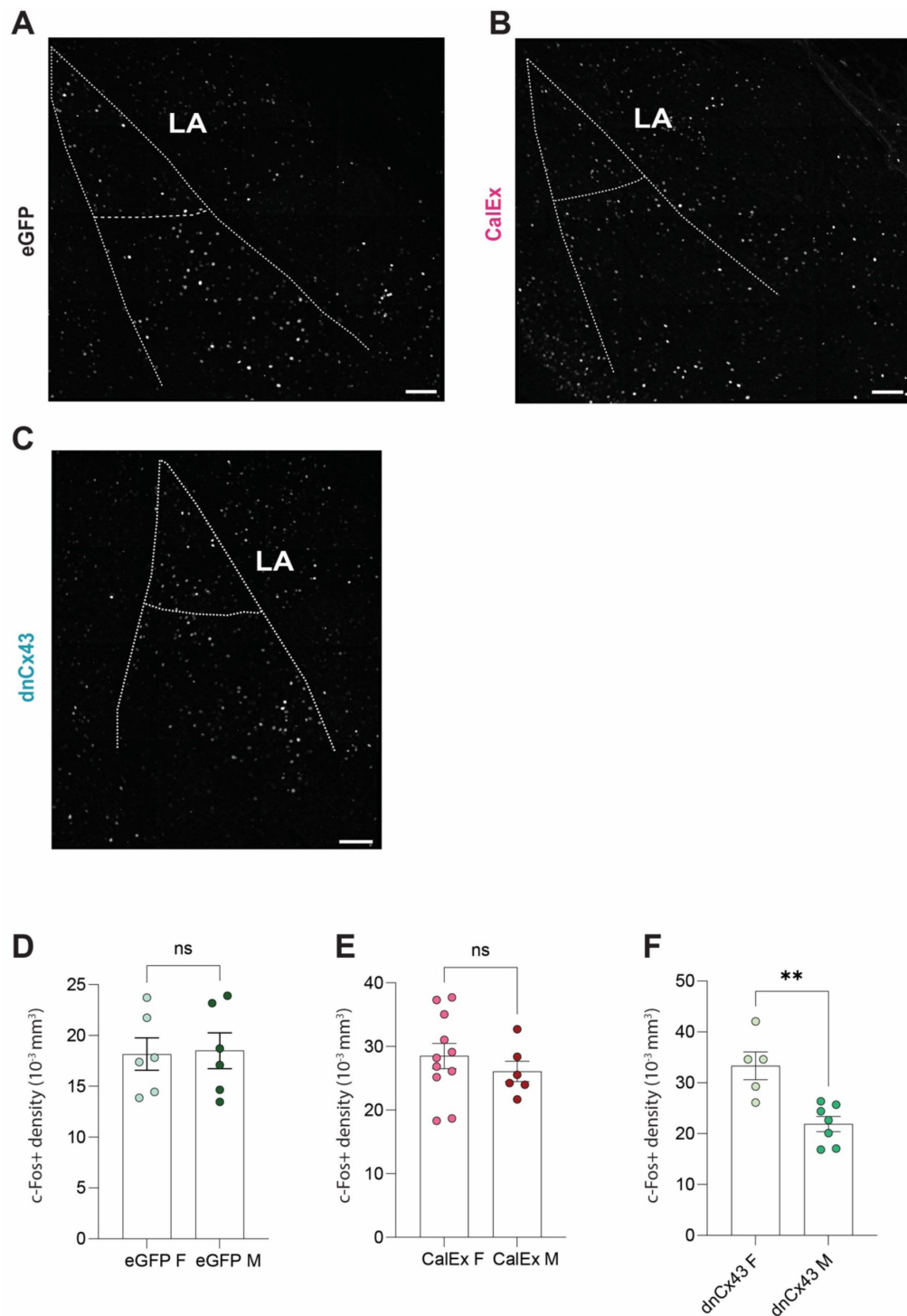


Figure 5 – Figure supplement 5

(A) Representative Z-stack projection of c-Fos staining in entire lateral and basolateral amygdala from an eGFP mice 90 minutes after conditioning. **(B)** Representative Z-stack projection of c-Fos staining in entire lateral and basolateral amygdala from a CalEx mice 90 minutes after conditioning. **(C)** Representative Z-stack projection of c-Fos staining in entire lateral and basolateral amygdala from a dnCx43 mice 90 minutes after conditioning. **(D)** No differences in lateral amygdala c-Fos density between male and female eGFP mice. $P=0.9372$, Male $n=6$ astrocytes, Female $n=6$ astrocytes. **(E)** No differences in lateral amygdala c-Fos density between male and female CalEx mice. $P=0.3502$, Male $n=6$ astrocytes, Female $n=11$ astrocytes. **(F)** dnCx43 leads to increased lateral amygdala c-Fos density in females compared to male counterparts. $P=0.0051$, Male $n=5$ astrocytes, Female $n=7$ astrocytes.

Figure 1	
1B	CORT at P45 P=0.01; Naïve=50.1±6.9 ng/ml, N=9; ELS=106.1±17.2 ng/ml, N=11; mean±s.e.m
1D	Time in periphery Naïve vs ELS Two-tailed unpaired t-test, t= 0.7542, P=0.4591; Naïve=241.8±10.33 seconds, N=9; ELS=230.3±10.27 seconds, N=14; mean±s.e.m
1E	Time in center Naïve vs ELS Two-tailed unpaired t-test, t= 0.7542, P=0.4591; Naïve= 58.17±10.33 seconds, N=9; ELS= 69.70±10.27 seconds, N=14; mean±s.e.m
1F	Total distance travelled in OF Naïve vs ELS Two-tailed unpaired t-test, t=2.128, P=0.0454; Naïve= 17.04±2.3 meters, N=9; ELS= 22.82±1.602 meters, N=14; mean±s.e.m
1H	Time in closed arms Naïve vs ELS Two-tailed unpaired t-test, t= 2.34, P= 0.0276; Naïve= 196.8± 9.763 seconds, N=16; ELS= 231.9± 11.16 seconds, N=11; mean±s.e.m
1I	Time in open arms Naïve vs ELS Two-tailed unpaired t-test, t= 2.34, P= 0.0276; Naïve= 103.2± 9.763 seconds, N=16; ELS= 68.13± 11.16 seconds, N=11; mean±s.e.m
1J	Total distance travelled in EPM Naïve vs ELS Two-tailed unpaired t-test, t= 2.167, P= 0.0439; Naïve= 8.167 ± 0.6223 meters, N=9; ELS= 11.03 ± 1.079 meters, N=11; mean±s.e.m
Figure 2	
2C	Fear conditioning CS- CS+ Naïve vs ELS Two-way ANOVA; effect of group: F (3, 360) = 64.48, P<0.0001; Tukey's multiple comparison test, Naïve CS- vs ELS CS-: P= 0.2124, Naïve CS+ vs ELS CS+: P= 0.2656, Naïve CS- vs Naïve CS+: P<0.0001, ELS CS- vs ELS CS+: P<0.0001,

Supplementary Table 1

	Naïve CS-= 18.34 ± 3.714 % Freezing, N=15 mice Naïve CS+= 37.18 ± 8.119 % Freezing, N=15 mice ELS CS-=13.55±3.491 % Freezing, N=17 mice ELS CS+=41.67±6.869 %Freezing, N=17 mice Mean ± s.e.m
2D	Discrimination test Naïve vs ELS Two-way ANOVA; effect of group: F (3, 360) = 28.32, P<0.0001; Tukey's multiple comparison test, Naïve CS- vs ELS CS-: P <0.0001, Naïve CS+ vs ELS CS+: P= 0.9197, Naïve CS- vs Naïve CS+: P<0.0001, ELS CS- vs ELS CS+: P=0.0009, Naïve CS-= 30.57±1.966% Freezing., N=15 mice Naïve CS+= 56.22±2.991% Freezing., N=15 mice ELS CS-=46.03±2.9 % Freezing., N=17 mice ELS CS+=58.34±2.2 % Freezing., N=17 mice Mean±s.e.m
2E	Discrimination index Naïve vs ELS Two-way ANOVA; effect of group F (1, 30) = 9.701, P=0.004; Naïve= 0.3629±0.06059; N=15 mice ELS=0.1275±0.03533, N=17 mice mean±s.e.m
2H	LTP Naïve vs ELS Unpaired t-test, P=0.0028; Naïve=142.1±8.720%, N=15 ELS=106.0±6.058, N=13 mean±s.e.m
2L	c-Fos density Naïve vs ELS Mann-Withney test, P<0.0001, U= 228, Naïve=17.42±0.9427 c-Fos density/10⁻³mm³, n=39 slices from N=10 mice ELS=25.45±1.044 c-Fos density/10⁻³mm³, n=37 slices from N=10 mice mean±s.e.m.
Figure 3	
3C	Nuclear:cytosolic GR ratio Naïve vs ELS P=0.026; Naïve=0.23±0.06 a.u., N=8 ELS=0.48±0.08 a.u., N=8 mean±s.e.m.
3F	GFAP P=0.02, unpaired t-test;

Supplementary Table 1 (continued)

	Naïve=1±0.17 a.u., N=11 ELS=0.5±0.1 a.u, N=12 mean±s.e.m.
3G	Cx43 P=0.008, unpaired t-test; Naïve=1±0.15 a.u., N=11 ELS=0.45±0.12 a.u., N=14 mean±s.e.m.
3H	GLT-1 P=0.8, unpaired t-test; Naïve=1±0.29 a.u., N=6 ELS=0.9±0.21 a.u. N=16 mean±s.e.m
Figure 4	
4F	Calcium event frequency Naïve vs ELS Two-tailed unpaired t test, t=4.894, P <0.0001 Naïve= 3.127±0.08503 Hz, n=25 astrocytes from N=8 animals ELS= 2.444 ±0.1106 Hz, n=25 astrocytes from N=8 animals
4H	Calcium event amplitude Naïve vs ELS Mann-Whitney test, P=0.0165 Naïve= 0.084±0.006979 au, n=22 astrocytes from N=8 animals ELS= 0.05818±0.005380 au, n=17 astrocytes from N=8 animals
4J	Calcium event area Naïve vs ELS Two-tailed unpaired t test, t=5.284, P <0.0001 Naïve= 4.642±0.3611 µm ² , n=25 astrocytes from N=8 animals ELS= 2.540±0.1670 µm ² , n=25 astrocytes from N=8 animals
4L	Calcium event rise time Naïve vs ELS Two-tailed unpaired t test, t=2.222, P=0.0311 Naïve= 4.782±0.1454 seconds, n=25 astrocytes from N=8 animals ELS= 5.299 ±0.1821 seconds, n=25 astrocytes from N=8 animals
4N	Calcium event duration Naïve vs ELS Two-tailed unpaired t test, t=0.7217, P=0.4740 Naïve= 8.987±0.2189 seconds, n=25 astrocytes from N=8 animals ELS= 9.3075 ±0.3869 seconds, n=25 astrocytes from N=8 animals
4P	Calcium event decay time Naïve vs ELS Mann-Whitney test, P=0.2164 Naïve= 7.136±0.3719 seconds, n=25 astrocytes from N=8 animals ELS= 8.147±0.6189 seconds, n=25 astrocytes from N=8 animals
Figure 5	
5D	Fear conditioning CS- CS+ eGFP vs CalEx vs dnCx43

Supplementary Table 1 (continued)

	Two-way ANOVA; effect of group: $F(5, 463) = 78.39$, $P < 0.0001$; Tukey's multiple comparison test, eGFP CS- vs dnCx43 CS-: $P = 0.1940$, eGFP CS- vs CalEx CS-: $P = 0.9758$, dnCx43 CS- vs CalEx CS-: $P = 0.7027$, eGFP CS+ vs dnCx43 CS+: $P = 0.9536$, eGFP CS+ vs CalEx CS+: $P = 0.6680$, dnCx43 CS+ vs CalEx CS+: $P = 0.9761$, eGFP CS- = 11.24 ± 1.875 % Freezing., N=12 mice CalEx CS- = 9.182 ± 2.247 % Freezing., N=11 mice dnCx43 CS- = 5.515 ± 1.936 % Freezing., N=17 mice eGFP CS+ = 39.33 ± 6.443 % Freezing., N=12 mice CalEx CS+ = 35.22 ± 7.464 % Freezing., N=11 mice dnCx43 CS+ = 37.14 ± 6.923 % Freezing., N=17 mice mean $\pm$ s.e.m
5E	Discrimination test eGFP vs CalEx vs dnCx43 freezing responses Two-way ANOVA; effect of group: $F(5, 457) = 25.29$, $P < 0.0001$; Tukey's multiple comparison test, eGFP CS- vs dnCx43 CS-: $P = 0.0293$, eGFP CS- vs CalEx CS-: $P < 0.0001$, dnCx43 vs CalEx: $P = 0.0005$, eGFP CS+ vs dnCx43 CS+: $P > 0.9999$, eGFP CS+ vs CalEx CS+: $P > 0.9999$, dnCx43 CS+ vs CalEx CS+: $P > 0.9999$ eGFP CS- = 28.73 ± 2.342 % Freezing., N=12 mice CalEx CS- = 53.00 ± 1.203 % freezing., N=11 mice dnCx43 CS- = 38.81 ± 1.957 % Freezing., N=17 mice eGFP CS+ = 57.79 ± 3.314 % Freezing., N=12 mice CalEx CS+ = 57.02 ± 4.219 % Freezing., N=11 mice dnCx43 CS+ = 57.32 ± 4.344 % Freezing., N=17 mice mean $\pm$ s.e.m.
5F	Discrimination index eGFP vs CalEx vs dnCx43 Two-way ANOVA; effect of group: $F(2, 226) = 12.24$, $P < 0.0001$; Tukey's multiple comparison test, eGFP vs dnCx43: $P = 0.0137$, eGFP vs CalEx: $P < 0.0001$, dnCx43 vs CalEx: $P = 0.0352$, eGFP DI = 0.3636 ± 0.03681; CalEx DI = 0.05023 ± 0.05221; dnCx43 DI = 0.1989 ± 0.03771;
5G	LTP eGFP vs CalEx vs dnCx43 Ordinary one-way ANOVA, group effect: $F(2, 31) = 4.221$, $P = 0.0239$;

Supplementary Table 1 (continued)

	Uncorrected Fisher's LSD, eGFP vs CalEx: P=0.0408, eGFP vs dnCx43: P=0.0207, CalEx vs dnCx43: P=0.8848, eGFP=139.0±17.70 % baseline., N=5; CalEx=100.3±7.485 % baseline., N=9; dnCx43=98.31±8.635 % baseline., N=16; mean± s.e.m
4L	c-Fos density eGFP vs CalEx vs dnCx43 Ordinary one-way ANOVA, group effect: F (2, 43) = 14.91, P<0.0001; Tukey's multiple comparison test, eGFP vs CalEx: P<0.0001, eGFP vs dnCx43: P=0.0005, CalEx vs dnCx43: P=0.8807 eGFP=17.97±0.8195 c-Fos density/10⁻³mm³, n=17 slices from N=5 mice; CalEx=27.66±1.398 c-Fos density/10⁻³mm³, n=17 from N=5 mice; dnCx43=26.65±2.173 c-Fos density/10⁻³mm³, n=12 slices from N=5 mice; mean±s.e.m
Figure supplements	
Figure 2 – Figure supplement 1	
S1A	Naïve Auditory Discriminative Fear Conditioning Female vs Male CONDITIONING Cs- Two-way ANOVA; effect of sex: F(1, 13)=1.20; P=0.291 Naïve Male=16.31±3.33; N=8 Naïve Female=23.9±5.12; N=7 Cs+/US Two-Way ANOVA; effect of sex: F(1, 13)=0.33; P=0.574 Naïve Male=36.11±8.54; N=8 Naïve Female=39.49±8.08; N=7
S1B	Naïve Auditory Discriminative Fear Conditioning Female vs Male TESTING Cs- Two-way ANOVA; effect of sex: F (1, 13) = 0.5316; P=0.4789 Naïve Male=27.97±2.18; N=8 Naïve Female=36.79±1.241; N=7 Cs+/US Two-Way ANOVA; effect of sex: F (1, 13) = 0.1696; P=0.6872 Naïve Male=54.75±2.58; N=8 Naïve Female=58.83±4.84; N=7

Supplementary Table 1 (continued)

S1C	Naïve Auditory Discriminative Fear Conditioning Female vs Male Discrimination Index Two-Way ANOVA; effect of sex: $F(1,13)=0.03$; $P=0.85$ Naïve Male=0.32 ± 0.16; $N=8$ Naïve Female=0.33 ± 0.06; $N=7$
S1D	ELS Auditory Discriminative Fear Conditioning Female vs Male CONDITIONING Cs- Two-way ANOVA; effect of sex: $F(1, 15) = 2.109$, $P=0.167$ ELS Male=17.69 ± 4.205; $N=8$ ELs Female=9.88 ± 3; $N=9$ Cs+/US Two-Way ANOVA; effect of sex: $F(1, 15) = 9.638$, $P=0.0072$ ELS Male=33.74 ± 6.28; $N=8$ ELS Female=48.71 ± 7.81; $N=9$
S1E	ELS Auditory Discriminative Fear Conditioning Female vs Male TESTING Cs- Two-way ANOVA; effect of sex: $F(1, 15) = 1.484$, $P=0.2419$ ELS Male=42.17 ± 2.997; $N=8$ ELs Female=50.39 ± 4.011; $N=9$ Cs+ Two-Way ANOVA; effect of sex: $F(1, 15) = 1.763$, $P=0.2041$ ELS Male=54.60 ± 2.054; $N=8$ ELS Female=62.54 ± 3.816; $N=9$
S1F	ELS Auditory Discriminative Fear Conditioning Female vs Male Discrimination Index Two-Way ANOVA; effect of sex: $F(1, 15) = 0.4431$; $P=0.515$ Naïve Male=0.123 ± 0.053; $N=8$ Naïve Female=0.149 ± 0.033; $N=9$
Figure 2 - Figure supplement 2	
S2A	Auditory Discriminative Fear Conditioning ELS vs. Naïve CONDITIONING Two-Way ANOVA; effect of group : $F(1,30)=2.613$; $P=0.116$; effect of tone type : $F(1, 30) = 53.36$, $P<0.0001$ CS-, Naïve vs ELS: $P= 0.2543$ CS+, Naïve vs Els : $P= 0.2851$

Supplementary Table 1 (continued)

	Naive, CS+ vs CS- : $P<0.0001$ ELS, CS+ vs CS- = $P<0.0001$ Naive CS- = 18.34 ± 3.71; N=15 Naive CS+/US = 37.18 ± 8.11; N=15 ELS CS- = 13.55 ± 3.49; N=17 ELS CS+/US = 41.67 ± 6.86; N=17
S2B	Auditory Discriminative Fear Testing ELS vs. Naïve TESTING Two-Way ANOVA; effect of group: $F(1,28)=0.0014$; $P=0.9701$; effect of tone type : $F(1, 32) = 43.23$, $P<0.0001$ CS-, Naive vs ELS: $P=0.0128$ CS+, Naive vs Els : $P=0.726$ Naive, CS+ vs CS- : $P=0.0001$ ELS, CS+ vs CS- = $P=0.0016$ Naive CS- = 30.57 ± 1.96; N=15 Naive CS+/US = 56.22 ± 2.99; N=15 ELS CS- = 46.03 ± 2.92; N=17 ELS CS+/US = 58.34 ± 2.26; N=17
S2F	c-Fos expression Females vs Males Naive Two-Tailed T-Test : $T=1.677$; $P=0.102$, Females = 19.74 ± 1.754, N=12 Males = 16.39 ± 1.08, N=27
S2G	c-Fos expression Females vs Males ELS Two-Tailed T-Test : $T=2.153$; $P=0.038$, Females = 23.25 ± 1.38, N=18 Males = 27.54 ± 1.429, N=19
Figure 3 – Figure supplement 1	
S3A	Normalized GFAP fluorescence Naïve ELS Male Female Two-Way ANOVA, effect of group $F(1, 29) = 8.915$, $P=0.0057$; effect of sex $F(1, 29) = 0.04720$, $P=0.8295$ Tukey's HSD: Male, Naïve vs ELS: $P=0.0582$ Female, Naïve vs ELS; $P=0.0323$ Naïve, Male vs Female: $P>0.999$ ELS, Male vs Female: $P=0.7501$ Female Naïve = 1.00 ± 0.2849, N=8

Supplementary Table 1 (continued)

	Male Naïve=1.00±0.2290, N=7 Female ELS=0.4052±0.0762, N=8 Male ELS=0.4859±0.1076, N=10
S3B	Cx43 fluorescence Naïve ELS Male Female Two-Way ANOVA, effect of group F (1, 28) = 11.66, P=0.0020; effect of sex F (1, 28) = 0.005276, P=0.9426
	Male, Naïve vs ELS: P=0.0253 Female, Naïve vs ELS; P=0.0201 Naïve, Male vs Female: P>0.999 ELS, Male vs Female: P=0.9189
	Female Naïve=1.00±0.09592, N=8 Male Naïve=1.00±0.2407, N=7 Female ELS=0.4452±0.0752, N=8 Male ELS=0.4683±0.1767, N=10
Figure 4 – Figure supplement 1	
S4A	Amplitude 2P Naïve ELS Male Female Two-Way ANOVA; effect of group F (1, 35) = 9.437; P=0.0041; effect of sex F (1, 35) = 1.904, P=0.1764
	Male, Naïve vs ELS: P=0.5478 Female, Naïve vs ELS; P=0.0058 Naïve, Male vs Female: P=0.0425 ELS, Male vs Female: P=0.9627
	Female Naïve=0.09±0.012, N=10 Male Naïve=0.071±0.005, N=12 Female ELS=0.0562±0.0111, N=7 Male ELS=0.0595±0.00539, N=10
S4D	Event Area 2P Naïve ELS Male Female Two-Way ANOVA; effect of group F (1, 46) = 41.50; P<0.0001; effect of sex F (1, 46) = 8.116, P=0.0065
	Male, Naïve vs ELS: P=0.0385 Female, Naïve vs ELS; P<0.0001 Naïve, Male vs Female: P=0.0002 ELS, Male vs Female: P=0.9702
	Female Naïve=5.806±0.562, N=11 Male Naïve=3.721±0.304, N=14

Supplementary Table 1 (continued)

	Female ELS=2.484±0.267, N=12 Male ELS=2.592±0.2151, N=13
S4G	Rise Time 2P Naïve ELS Male Female Two-Way ANOVA; effect of group $F(1, 46) = 9.458$; $P < 0.0035$; effect of sex $F(1, 46) = 1.152$, $P = 0.2886$ Tukey's HSD: Male, Naïve vs ELS: $P = 0.7971$ Female, Naïve vs ELS; $P < 0.0001$ Naïve, Male vs Female: $P = 0.0021$ ELS, Male vs Female: $P = 0.1002$ Female Naïve=4.192±0.1243, N=11 Male Naïve=5.195±0.1388, N=14 Female ELS=5.594±0.1575, N=12 Male ELS=5.027±0.3061, N=13
S4J	Frequency 2P Naïve ELS Male Female Two-Way ANOVA; effect of group $F(1, 46) = 22.84$; $P < 0.0001$; effect of sex $F(1, 46) = 0.01429$, $P = 0.9054$ Male, Naïve vs ELS: $P < 0.0001$ Female, Naïve vs ELS; $P = 0.0419$ Naïve, Male vs Female: $P = 0.2053$ ELS, Male vs Female: $P = 0.2671$ Female Naïve=2.985±0.132, N=11 Male Naïve=3.288±0.108, N=14 Female ELS=2.558±0.129, N=12 Male ELS=2.339±0.175, N=13
S4M	Duration 2P Naïve ELS Male Female Two-Way ANOVA; effect of group $F(1, 46) = 0.7290$; $P < 0.3976$; effect of sex $F(1, 46) = 0.2611$, $P = 0.6118$ Male, Naïve vs ELS: $P = 0.8877$ Female, Naïve vs ELS; $P = 0.6991$ Naïve, Male vs Female: $P = 0.8304$ ELS, Male vs Female: $P = 0.9806$ Female Naïve=8.741±0.3951, N=11

Supplementary Table 1 (continued)

	Male Naïve=9.093±0.2480, N=14 Female ELS=9.249±0.5234, N=12 Male ELS=9.362±0.5861, N=13
S4P	Decay Time 2P Naïve ELS Male Female Two-Way ANOVA; effect of group $F(1, 46) = 2.459$; $P < 0.1237$; effect of sex $F(1, 46) = 0.08601$, $P = 0.7706$ Tukey's HSD: Male, Naïve vs ELS: $P = 0.9997$ Female, Naïve vs ELS; $P < 0.0718$ Naïve, Male vs Female: $P = 0.5942$ ELS, Male vs Female: $P = 0.3383$ Female Naïve=6.570±0.5769, N=11 Male Naïve=7.512±0.4867, N=14 Female ELS=8.858±0.9071, N=12 Male ELS=7.492±0.8382, N=13
Figure 5 – Figure supplement 2	
S6F	Event frequency CalEx Male vs Female Mann-Whitney U test, $U = 2$, $P = 0.0087$ CalEx- = 2.571±0.4405, N=6 CalEx+ 1.264±0.1027, N=6
S6G	Amplitude CalEx Male vs Female Mann-Whitney U test, $U = 2$, $P = 0.0411$ CalEx- = 1.432±0.04066, N=6 CalEx+ = 0.0705±0.0063, N=6
S6H	Event area CalEx Male vs Female Mann-Whitney U test, $U = 6$, $P = 0.0649$ CalEx- = 4.428±1.275, N=6 CalEx+ = 2.071±0.2492, N=6
S6I	Duration CalEx Male vs Female Mann-Whitney U test, $U = 9$, $P = 0.1797$ CalEx- = 6.295±0.4358, N=6 CalEx+ = 5.480±0.5079, N=6
S6J	Rise Time CalEx Male vs Female Mann-Whitney U test, $U = 11$, $P = 0.3095$

Supplementary Table 1 (continued)

	CalEx- = 5.066±1.060, N=6 CalEx+ = 3.656±0.3663, N=6
S6K	Decay Time CalEx Male vs Female Mann-Whitney U test, U=14, P=0.5887 CalEx- = 5.136±0.5837, N=6 CalEx+ = 4.723±0.5591, N=6
Figure 5 – Figure supplement 3	
S7A	EPM – Time in closed arms eGFP CalEx dnCx43 Kruskal-Wallis Test H=1.465, P=0.4807 eGFP=225.1±15.15, N=7 CalEx=237.9±5.899, N=9 dnCx43=242.9±9.076, N=9
S7B	EPM – Time in open arms eGFP CalEx dnCx43 Kruskal-Wallis Test H=1.465, P=0.4807 eGFP=74.88±15.15, N=7 CalEx=62.14±5.899, N=9 dnCx43=57.06±9.076, N=9
S7C	EPM – Distance Traveled eGFP CalEx dnCx43 One-Way ANOVA, main effect of condition F (DFn, DFd); P=0.3231 eGFP=11.38±1.379, N=7 CalEx=9.421±0.7355, N=9 dnCx43=9.898±0.6352, N=9
S7D	Open Field – Time in Periphery eGFP CalEx dnCx43 Kruskal-Wallis Test H=1.528, P=0.4658 eGFP=36.95±8.727, N=6 CalEx=50.05±8.165, N=11 dnCx43=37.73±6.460, N=12
S7E	Open Field – Time in Center eGFP CalEx dnCx43 Kruskal-Wallis Test H=1.528, P=0.4658 eGFP=263.1±8.727, N=6 CalEx=249.9±8.165, N=11 dnCx43=262.3±6.460, N=12

Supplementary Table 1 (continued)

S7F	Open Field – Distance Traveled eGFP CalEx dnCx43 Kruskal-Wallis Test $H=4.282$, $P=0.1176$ eGFP=22.12 ± 5.799, $N=6$ CalEx=15.68 ± 1.292, $N=11$ dnCx43=17.42 ± 1.666, $N=12$
Figure 5 – Figure supplement 4	
S8A	Auditory Discriminative Fear Conditioning eGFP vs CalEx vs dnCx43 CONDITIONING Two-Way ANOVA; effect of group : $F(2, 74) = 1.406$; $P=0.2516$; effect of tone type : $F(1, 74) = 159.1$, $P<0.0001$ eGFP, CS- vs CS+/US, $P<0.0001$ CalEx, CS- vs CS+/US, $P<0.0001$ dnCx43, CS- vs CS+/US, $P<0.0001$ CS-, eGFP vs CalEx, $P=0.944$ CS-, eGFP vs dnCx43, $P=0.3372$ CS-, dnCx43 vs CalEx, $P=0.7121$ CS+/US, eGFP vs CalEx, $P=0.6879$ CS+/US, eGFP vs dnCx43, $P=0.8030$ CS+/US, dnCx43 vs CalEx, $P=0.9891$ CS- eGFP = 11.24 ± 2.754, $N=12$ CS- CalEx = 53 ± 5.859, $N=11$ CS- dnCx43 = 38.59 ± 3.374, $N=17$ CS+/US eGFP = 39.33 ± 4.552, $N=12$ CS+/US CalEx = 35.22 ± 3.492, $N=11$ CS+/US dnCx43 = 36.3 ± 2.07, $N=17$
S8B	Auditory Discriminative Fear Conditioning dGFP vs CalEx vs dnCx43 TESTING Two-Way ANOVA; effect of group : $F(2, 74) = 3.612$; $P=0.0318$; effect of tone type : $F(1, 74) = 23.83$, $P<0.0001$ eGFP, CS- vs CS+, $P<0.0001$ CalEx, CS- vs CS+, $P=0.534$ dnCx43, CS- vs CS+, $P=0.0014$ CS-, eGFP vs CalEx, $P=0.0007$ CS-, eGFP vs dnCx43, $P=0.02383$ CS-, dnCx43 vs CalEx, $P=0.0467$ CS+/US, eGFP vs CalEx, $P=0.9991$ CS+/US, eGFP vs dnCx43, $P=0.9779$ CS+/US, dnCx43 vs CalEx, $P=0.9949$ CS- eGFP = 28.73 ± 3.826, $N=12$

Supplementary Table 1 (continued)

	CS- CalEx = 53.0±5.859, N=11 CS- dnCx43 = 38.59±3.374, N=17 CS+/US eGFP = 57.79±4.558, N=12 CS+/US CalEx = 57.02±4.644, N=11 CS+/US dnCx43 = 55.74±3.260, N=17
S8C	Auditory Discriminative Fear Conditioning – Discrimination Index CalEx Male vs Female Two-Way ANOVA, main effect of sex $F(5, 54) = 3.726$; $P=0.0057$
S8D	Auditory Discriminative Fear Conditioning – Discrimination Index dnCx43 Male vs Female Two-Way ANOVA, main effect of sex $F(5, 90) = 1.435$; $P=0.2195$
Figure 5 – Figure supplement 5	
S9D	c-Fos density, eGFP Male vs Female Mann-Whitney test, $U=17$, $P=0.9372$ Female = 18.16±1.599, N=6 Male = 18.50±1.762, N=6
S9E	c-Fos density, CalEx Male vs Female Mann-Whitney test, $U=23$, $P=0.3502$ Female = 28.51±1.982, N=11 Male = 26.10±1.601, N=6
S9F	c-Fos density, dnCx43 Male vs Female Mann-Whitney test, $U=1$, $P=0.0051$ Female = 33.32±2.722, N=5 Male = 21.88±1.488, N=7

Supplementary Table 1 (continued)

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M.G. I.I.A., L.R.D-H, and C.M-R designed the study. M.G., I.I.A., O.G., J.L-A, J.V. and C.M-R. carried out behavioural experiments, analysing the subsequent data. M.G. and L.R.D-H. carried out immunostaining and analysis of subsequent data. L.R.D-H, M.D., B.R. took blood samples and performed ELISA experiments. M.G., L.R.D-H., O.G., and A.B. carried out injection of viral vectors. M.G. and A.B. carried out electrophysiological experiments and analysis of subsequent data. M.G. carried out calcium imaging. S.P. maintained and generated mouse colonies. M.G. L.R.D-H, A.B., and C.M-R wrote the manuscript. All authors contributed to generating ELS mice and approved the final version of the manuscript.

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Reviewer #1 (Public review):

Summary:

The manuscript asks the question of whether astrocytes contribute to behavioral deficits triggered by early life stress. This question is tested by experiments that monitor the effects of early life stress on anxiety-like behaviors, long-term potentiation in the lateral amygdala, and immunohistochemistry of astrocyte-specific (GFAP, Cx43, GLT-1) and general activity (c-Fos) markers. Secondarily, astrocyte activity in the lateral amygdala is impaired by viruses that suppress gap-junction coupling or reduce astrocyte Ca^{2+} followed by behavioral, synaptic plasticity, and c-Fos staining. Early life stress is found to reduce expression of GFAP, Cx43 and induce translocation of the glucocorticoid receptor to astrocytic nuclei. Both early life stress and astrocyte manipulations are found to result in generalization of fear to neutral auditory cues. All of the experiments are done well with appropriate statistics and control groups. The manuscript is very well-written and the data are presented clearly. The authors' conclusion that lateral amygdala astrocytes regulate amygdala-dependent behaviors is strongly supported by the data as is the conclusion that cellular and behavioral outcomes provoked by early life stress are similar to the outcomes provoked by astrocyte dysfunction. However, the extent to which early life stress requires astrocytes to generate these outcomes remains open to debate.

Strengths:

A strong combination of behavioral, electrophysiology, and immunostaining approaches is utilized and possible sex-differences in behavioral data are considered. The experiments clearly demonstrate that disruption of astrocyte networks or reduction of astrocyte Ca^{2+} provoke generalization of fear and impair long-term potentiation in lateral amygdala. The provocative finding that astrocyte dysfunction accounts for a subset of behavioral effects of early life stress (e.g. not elevated plus or distance traveled observations) is also perceived as a strength.

Weaknesses:

The main weakness is absence of direct evidence that behavioral and neuronal plasticity after early life stress can be attributed to astrocytes. It remains unknown what would happen if astrocyte activity were disrupted concurrently with early life stress or if changes in astrocyte Ca^{2+} could attenuate early life stress outcomes. As is, the only presented evidence that early life stress involves astrocytes is nuclear translocation of GR and downregulation of GFAP and Cx43 in Figure 3 which may or may not cause the reported astrocyte activity changes.

<https://doi.org/10.7554/eLife.99988.2.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, Guayasamin et al. show that early-life stress (ELS) can induce a shift in fear generalisation in mice. They took advantage of a fear conditioning paradigm followed by a discrimination test and complement learning and memory findings with measurements for anxiety-like behaviors. Next, astrocytic dysfunction in the lateral amygdala was investigated at the cellular level by combining staining for c-Fos with astrocyte-related proteins. Changes in excitatory neurotransmission were observed in acute brain slices after ELS suggesting impaired communication between neurons and astrocytes. To confirm causality of astrocytic-neuronal dysfunction in behavioral changes, viral manipulations were performed in unstressed mice. Occlusion of functional coupling with a dominant negative construct for gap junction connexin 43 or reduction in astrocytic calcium with CalEx mimicked the behavioral changes observed after ELS suggesting that dysfunction of the astrocytic network underlies ELS-induced memory impairments.

Strengths:

Overall, this well written manuscript highlights a key role for astrocytes in regulating stress-induced behavioral and synaptic deficits in the lateral amygdala in the context of ELS. Results are innovative, and methodological approaches relevant to decipher the role of astrocytes in behaviors. As mentioned by the authors, non-neuronal cells are receiving increasing attention in the neuroscience, stress and psychiatry fields.

Weaknesses:

I did have several suggestions and comments that were addressed during the review process. I believe that it improved clarity and will increase the impact of the work.

<https://doi.org/10.7554/eLife.99988.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors show that ELS induces a number of brain and behavioral changes in the adult lateral amygdala. These changes include enduring astrocytic dysfunction, and inducing astrocytic dysfunction via genetic interventions is sufficient to phenocopy the behavioral and neural phenotypes suggesting astrocyte dysfunction may play a causal role in ELS-associated pathologies.

Strengths:

A strength is the shift in focus to astrocytes to understand how ELS alters adult behavior.

Weaknesses:

The mechanistic links between some of the correlates - altered astrocytic function, changes in neural excitability and synaptic plasticity in the lateral amygdala and behavior - are underdeveloped.

Comments on revisions:

The authors have significantly improved the paper with the addition of new experimental data, analyses, and textual changes.

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

Early-life adversity or stress can enhance stress susceptibility by causing changes in emotion, cognition, and reward-seeking behaviors. This important manuscript highlights the involvement of lateral amygdala astrocytes in fear generalization and the associated synaptic plasticity, which are parallel to the effects of early life stress. With an elegant combination of behavioral models, morphological and functional assessments using immunostaining, electrophysiology, and viral-mediated loss-of-function approaches, the authors provide solid correlational and causal evidence that is consistent with the hypothesis that early life stress produces neural and behavioral dysfunction via perturbing lateral amygdala astrocytic function.

We would like to thank the authors and editors for taking the time to review our work, and re-review it now. Also, we are grateful for this very positive assessment of our work. In this revised manuscript we made a strong effort to address comments made by all reviewers, providing clarification where required and new data to our manuscript in order to further support our observations.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The manuscript asks the question of whether astrocytes contribute to behavioral deficits triggered by early life stress. This question is tested by experiments that monitor the effects of early life stress on anxiety-like behaviors, long-term potentiation in the lateral amygdala, and immunohistochemistry of astrocyte-specific (GFAP, Cx43, GLT-1) and general activity (c-Fos) markers. Secondly, astrocyte activity in the lateral amygdala is impaired by viruses that suppress gap-junction coupling or reduce astrocyte Ca²⁺ followed by behavioral, synaptic plasticity, and c-Fos staining. Early life stress is found to reduce the expression of GFAP and Cx43 and to induce translocation of the glucocorticoid receptor to astrocytic nuclei. Both early life stress and astrocyte manipulations are found to result in the generalization of fear to neutral auditory cues. All of the experiments are done well with appropriate statistics and control groups. The manuscript is very well-written and the data are presented clearly. The authors' conclusion that lateral amygdala astrocytes regulate amygdala-dependent behaviors is strongly supported by the data. However, the extent to which astrocytes contribute to behavioral and neuronal consequences of early life stress remains open to debate.

Strengths:

A strong combination of behavioral, electrophysiology, and immunostaining approaches is utilized and possible sex differences in behavioral data are considered. The experiments clearly demonstrate that disruption of astrocyte networks or reduction of astrocyte Ca²⁺ provokes generalization of fear and impairs long-term potentiation in the lateral amygdala. The provocative finding that astrocyte dysfunction accounts for a subset of behavioral effects of early life stress (e.g. not elevated plus or distance traveled observations) is also perceived as a strength.

Weaknesses:

The main weakness is the absence of more direct evidence that behavioral and neuronal plasticity after early life stress can be attributed to astrocytes. It remains unknown what would happen if astrocyte activity were disrupted concurrently with early life stress or if the facilitation of astrocyte Ca^{2+} would attenuate early life stress outcomes. As is, the only evidence that early life stress involves astrocytes is nuclear translocation of GR and downregulation of GFAP and Cx43 in Figure 3 which may or may not provoke astrocyte Ca^{2+} or astrocyte network activity changes.

We would like to thank the reviewer for their constructive feedback on our work. In the revised version we have added new experiments that further support a role of astrocytes in ELS-induced behavioural dysfunction. Specifically, we carried out two-photon calcium imaging in lateral amygdala astrocytes using viral overexpression of membrane tethered GCaMP6f. These experiments revealed a decrease in astrocyte calcium activity following ELS (Figure 4). Interestingly these data also showed an important number of sex differences (Figure 4 - Figure supplement 1).

These new data allow us to strengthen the link between ELS-induced astrocyte hypofunction and behavioural changes. Indeed, we validated the impact of CalEx on astrocyte calcium activity in the lateral amygdala, again using two-photon microscopy, and show that CalEx resulted in an astrocyte calcium signature that very closely resembled that of ELS, i.e. reduced frequency and amplitude of events (Figure 5 - Figure supplement 2). As such, we feel like these data, while still correlative in nature, strengthen our findings and conclusion that astrocyte dysfunction alone is sufficient to recapitulate the effects of stress on excitability, synaptic function, and behaviour.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Guayasamin et al. show that early-life stress (ELS) can induce a shift in fear generalisation in mice. They took advantage of a fear conditioning paradigm followed by a discrimination test and complemented learning and memory findings with measurements for anxiety-like behaviors. Next, astrocytic dysfunction in the lateral amygdala was investigated at the cellular level by combining staining for c-Fos with astrocyte-related proteins. Changes in excitatory neurotransmission were observed in acute brain slices after ELS suggesting impaired communication between neurons and astrocytes. To confirm the causality of astrocytic-neuronal dysfunction in behavioral changes, viral manipulations were performed in unstressed mice. Occlusion of functional coupling with a dominant negative construct for gap junction connexin 43 or reduction in astrocytic calcium with CalEx mimicked the behavioral changes observed after ELS suggesting that dysfunction of the astrocytic network underlies ELS-induced memory impairments.

Strengths:

Overall, this well-written manuscript highlights a key role for astrocytes in regulating stress-induced behavioral and synaptic deficits in the lateral amygdala in the context of ELS. Results are innovative, and methodological approaches relevant to decipher the role of astrocytes in behaviors. As mentioned by the authors, non-neuronal cells are receiving increasing attention in the neuroscience, stress, and psychiatry fields.

Weaknesses:

I do have several suggestions and comments to address that I believe will improve the clarity and impact of the work. For example, there is currently a lack of information on the timeline for behavioral experiments, tissue collection, etc.

We thank the reviewer for their kind comments and constructive feedback on our manuscript. We agree that certain aspects could have been made more clear and we have revised the manuscript and figures to be more explicit regarding timelines. Including the addition of timelines on figures and improved clarity in the text where possible. We have also addressed the private comments provided by the reviewers alluded to in this public review.

Reviewer #3 (Public Review):

Summary

The authors show that ELS induces a number of brain and behavioral changes in the adult lateral amygdala. These changes include enduring astrocytic dysfunction, and inducing astrocytic dysfunction via genetic interventions is sufficient to phenocopy the behavioral and neural phenotypes. This suggests that astrocyte dysfunction may play a causal role in ELS-associated pathologies.

Strengths:

A strength is the shift in focus to astrocytes to understand how ELS alters adult behavior.

Weaknesses:

The mechanistic links between some of the correlates - altered astrocytic function, changes in neural excitability, and synaptic plasticity in the lateral amygdala and behaviour - are underdeveloped.

We thank the reviewer for their comments. We are happy that they found our shift in focus towards astrocytes to be a strength of our work. Regarding mechanistic links being underdeveloped, we have attempted to address this by placing more effort into understanding the functional changes in astrocytes and how this relates to behaviour.

To address this comment we have used two-photon calcium imaging to quantify the impact of ELS on astrocyte calcium activity. As such, the revised manuscript contains several new figures including a detailed characterisation of the effects of ELS on astrocyte calcium activity (Figure 4), including sex differences in naive and the effects of stress (Figure 4 - Figure supplement 1), and an important validation of the impact of CalEx on astrocyte calcium activity. CalEx mirrors the impact of stress on astrocyte calcium activity reducing the frequency and amplitude of individual events (Figure 5 - Figure supplement 2).

Considering the strong overlap of the effects of ELS and CalEx on synapses, excitability, behaviour, and now astrocyte calcium activity, we hope that this added detail addresses some of the points highlighted by the reviewer.

Recommendations for the authors:

The reviewers all agree on one major issue for the authors to address. There is a bit of a lack of mechanistic linking between the astrocyte function and the early life stress and these data are more correlational than causal in nature. This could either be addressed by scaling back the data interpretation and title to be more reflective of the data at hand or if the authors would consider, doing the causal experiment of examining the manipulation of astrocyte activity following early life stress to see if this does influence the phenotype.

We agree with reviewers on this issue and realise that we have overstated our findings somewhat. As an immediate fix, suggested by reviewers, we have changed the title to more closely align with our data stating that astrocyte dysfunction is “associated with” rather than “induces” as well as adjusting our interpretations.

In addition to this one major comment, there are a list of minor comments that the authors should consider to improve the manuscript.

(1) A major caveat is the lack of information on the timeline for behavioral experiments, tissue collection, etc. The authors mention "Mice between ages P45-70" but considering the developmental changes occurring between late adolescence and young adulthood, I recommend adding timelines on all Figures clearly indicating when behavioral tests were performed, and tissue collected for electrophysiology or immunostaining. With corticosterone (CORT) back at baseline at P70 vs a difference observed at P45 was this time point favored? It should be clarified throughout.

We apologise for the lack of clarity on this and have added more timelines on figures.

The age range favoured (p45-p70), relates to adolescence a time when latent psychiatric disorders tend to manifest in humans following early-life adversity. We have clarified this choice in the text.

(2) Given the transient increase in corticosterone levels in early-life stress mice, peaking at P45 and declining to control levels by P70, it would be informative to know whether the reported behavioral and synaptic changes differ within this time window. This may not be doable in the current approach, but this should be addressed nonetheless. Furthermore, it wasn't clear why the increase in blood corticosterone was delayed. Was this expected? How does this relate to earlier work? Wouldn't it be expected to be elevated at P17 (end of ELS period)?

We agree that this observation was very unexpected. Initially, we expected CORT to be elevated at P17, end of ELS period. We believe that low CORT levels during the ELS paradigm can be attributed to this paradigm coinciding with the stress hyporesponsive period (SHRP) which in rodents lasts until roughly postnatal day 14. During this period, mild stressors fail to elicit CORT responses. Considering our ELS paradigm lasts from P10-P17, there is a significant overlap with the SHRP.

This point is now included in the discussion with several citations regarding this biological phenomenon, as well as other studies that report similar findings to our own, i.e. a delayed increase in blood corticosterone levels following early-life stress.

(3) It is mentioned that behavioral tests were performed in both sexes with no sex differences observed. Were animals of both sexes also included in other experiments (ephys, immunostaining, blood CORT analysis)? Behavioral outcomes could be the same but underlying biological processes different. This is a topic that should be discussed. Identification of males vs females on graphs would be helpful.

We apologise for not having provided this data in the previous version of the manuscript. In the revised manuscript we provide analysis of sex differences for our initial behavioural observations (Figure 2 - Figure supplement 1), c-Fos (Figure 2 - Figure supplement 2), for GFAP and Cx43 (Figure 3 - Figure supplement 1), calcium signalling (Figure 4 - Figure supplement 1), and for CalEx and dnCx43 experiments across behaviour (Figure 5 - Figure supplement 4) and c-Fos (Figure 5 - Figure supplement 5).

(4) How long-lasting are the generalization phenotypes? Do they outlast the transient increase in blood corticosterone? Showing this would provide a more solid foundation for future explorations.

The reviewers raise a very important point. It remains unclear as to how long these effects last and this is something we are keen to address in future studies, with careful experiments designed to explicitly test this question, as well as subsequent questions regarding whether long-lasting effects are due to impaired brain development or whether these effects emerge due to CORT changes, or other changes, or a combination of them all?

As an aside, an additional manuscript from our lab (Depaauw-Holt et al. 2024 bioRxiv) which uses the same stressor but focuses on distinct brain regions and behaviours uses a prolonged time window in which the effects of stress are readily observable all the way to P90.

So while we do provide the answers in this work, it is a really great idea that we would like to follow up subsequently.

(5) With the ELS-induced change in locomotion, I would recommend presenting open field (center, periphery) and elevated plus maze (open, closed arms) data independently. It could also be interesting to analyze corner time in the open field as well as center time in the elevated plus maze.

We now provide data for the open field and elevated plus maze as requested. Our findings remain unchanged, but we agree with the reviewer that this way of representing the data is more clear.

(6) For Figure 2C, the ideal stats would be an ANOVA with CS (+/-) as a within-subject variable and treatment (naive/ELS) as a between-subjects variable. Then the best support for the generalization claim would be a CS x treatment interaction. I encourage the authors to do these stats. I note that this point is mitigated by the discrimination analysis presented in 2D (where they compare naive and ELS groups directly).

We have carried out the analysis as requested and these data further support the notion of fear generalisation in ELS mice (Figure 2 - Figure supplement 2A, B). Additionally, the analyses are included in a supplementary table. We hope that we have understood correctly, and this figure accurately reflects the reviewer's suggestion.

(7) In Figure 2H, why not evaluate c-Fos levels after the discrimination test which is the main behavioral outcome? This statement in the Discussion should be modified if, as per my understanding, c-Fos was measured after the fear paradigm only "We find that both ELS and astrocyte dysfunction both enhance neuronal excitability, assessed by local c-Fos staining in the lateral amygdala following auditory discriminative fear conditioning. One interpretation of these data is that astrocytes might tune engram formation, with astrocyte dysfunction, genetically or after stress, increasing c-Fos expression resulting in a loss of specificity of the memory trace and generalisation of fear."

We agree that further evaluation of c-Fos levels following the discrimination test would be insightful. We honestly did not consider this time point in our initial experimental design, as we considered previous reports in the literature that investigated how the numbers of cells recruited to the engram (c-Fos density) could influence memory accuracy at a later time point. As such, investigating c-Fos levels following training was our initial target. We have modified the text to be more explicit in our experimental approach.

This is nevertheless a fascinating point that we are keen to pursue in future studies.

(8) Some thoughts on why dnCx43 suppression of astrocyte network activity is less effective at inducing fear generalization than CalEx suppression of astrocyte Ca²⁺ are warranted. One might predict that both manipulations should result in similar effects, as seen in fEPSP and cFos data in Figure 4.

We agree that this is an interesting observation and the fact we did not observe the same behavioural phenotype despite fEPSP and c-Fos data to be the same is puzzling.

Nevertheless, we do see increased fear generalisation in both dnCx43 and CalEx. We hypothesise that CalEx had a more profound effect due to the wide range of processes that are presumably affected by reduced astrocyte calcium activity, whereas blocking gap junction channels still leaves a large number of astrocyte functions intact.

Overall, our conclusion is that behaviour is a more sensitive assay compared to the cellular phenotypes, which highlights the importance of answering these questions from multiple angles.

(9) Ideally changes in functional coupling following the dnCx43 manipulation) should be shown here (line 169).

We, unfortunately, did not directly evaluate functional coupling in dnCx43 mice in this manuscript. This would have been a useful experiment, but we rely on our previous data where we extensively characterised this tool (Murphy-Royal et al. 2020 Nat Comms).

(10) It would be relevant to perform c-Fos staining with markers for astrocytes or neuronal cells. Is an increase in activity expected for both cell types?

This is a fascinating question, given recent work on this topic showing that astrocytes can indeed express c-Fos and may be recruited into engrams. We analysed our existing tissue, we found that indeed astrocytes were labelled with c-Fos following our behavioural conditioning paradigm. Our data align with recent reports, and we demonstrate a small percentage of astrocytes expressing c-Fos (Figure 2 - figure supplement 3). This modest number of astrocytes expressing c-Fos is discussed in the text and placed into context of very recent papers that have been published since our submission to eLife.

(11) Were the same mice subjected to behavior analysis than immunostaining?

We generated separate cohorts of mice for immunostaining and behaviour, and have made this more clear in the text.

(12) Language describing learning paradigm. The CS+ (line 73) isn't in itself aversive (and shouldn't be described as such). It acquires that value after pairing with the US (which is aversive).

We agree that this is poorly worded and have modified the text from “aversive cue” to “conditioned cue”.

(13) It is hard to appreciate the glucocorticoid receptor translocation with the images provided. Would it be possible to increase magnification or at least, provide small inserts at higher magnification?

We have re-imaged our brain sections to get more detailed images. These are provided in revised manuscript (Figure 3)

(14) For the viral injection experiment, for how long is the virus expressed before running behavior/recording/c-Fos staining? Is the age of the tested mice the same as Figures 1-3 or they were injected at P45 and tested weeks later?

We age-matched all mice for all experiments and tried to keep our experimental window as tight as possible (p45-70). All mice were injected at P25-30 in order to meet the experimental time window. To be more precise we have added timelines on all figures.

(15) A validation of the virus is missing to confirm the reduction of Cx43 expression at mRNA and protein levels when compared to controls. A reference is provided but to my understanding age of the animals might be different.

Here, I believe the reviewer is referring to dnCx43. In this experiment we used a viral approach to overexpress a non-functional connexin 43 protein (Murphy-Royal et al. 2020 Nat Comms). As such, a PCR or immuno against this protein would be expected to reveal higher expression levels. We have tried to clarify this approach in the text.

It is true that we did not fully characterise this tool in the lateral amygdala which would have been useful but considering our extensive experience with this tool and in its development with our collaborators Baljit Khakh, Randy Stout, David Spray (see Murphy-Royal et al. 2020) we are confident in these data, despite the limitation of validation in this manuscript.

(16) Same comment for the CalEx, a validation would be appreciated. Based on Yu et al. could a GCaMP6f virus be more appropriate as control?

We agree this is an important experiment as our lab has not fully validated this tool in house (compared to dnCx43, which we previously validated).

Importantly, we now have the capacity to do these experiments. Until very recently our two-photon microscope was not fully functional due to dodgy PMTs sent from the company we purchased our equipment from... Troubleshooting this issue took many months before we were convinced that we were not at fault and that the problem was the equipment.

As such, mice were injected with both a membrane tethered GCaMP6f under the control of the short GFAP promoter - AAV2/5-gfaABC1D-lck-GCaMP6f and CalEx - AAV2/5-gfaABC1D-hPMCA2w/b-mCherry. Using this approach we were able to record calcium activity from CalEx positive and CalEx negative astrocytes in the same tissue (Figure 5 - figure supplement 2).

We report that this approach does indeed reduce astrocyte calcium but does not entirely eliminate it. In fact, CalEx expressing astrocytes displayed similar calcium activity dynamics to that we observed following ELS. Together, this further strengthens our rationale to use CalEx in order to mimic the effects of stress on astrocytes, and determine downstream effects on excitability, synapses, and behaviour.

(17) Have previous studies found ELS--> generalization phenotypes in adulthood? If so, these should be discussed in more detail. If not, perhaps this point can be made more explicit.

This is a great point. After looking deeper into the literature in more depth we found an example of this in which ELS resulted in context fear generalisation in adult rats. This work is cited in the discussion in the context of our findings.

(18) A paper by Krugers et al (Biol Psychiatry 2020) seems especially relevant (glucocorticoids, fear generalization, engram size) and should be discussed.

Thank you for bringing this work to our attention. This is certainly important work that we had unfortunately overlooked. We have added a citation and discussed the manuscript Lesuis et al. Biol. Psychiatry 2021, which contains the data discussed in the conference proceeding by Krugers et al. Biol. Psychiatry 2020.

Additionally, we added another great manuscript by Lesuis et al. recently published in *Cell* in which they investigated the cellular mechanisms by which acute stress results in fear generalisation via endocannabinoids.

(19) Minor text revisions are necessary at lines 101 and 264 as well as p.5, line 58: "ratio" and p.10, line 128: "region of interest".

Thank you for pointing out these typos and errors. We have corrected them.

Editor's note:

Should you choose to revise your manuscript, please include full statistical reporting including exact p-values wherever possible alongside the summary statistics (test statistic and df) and 95% confidence intervals. These should be reported for all key questions and not only when the p-value is less than 0.05 in the main manuscript.

<https://doi.org/10.7554/eLife.99988.2.sa0>